

Transported Complier Effects with Weak First Stages

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Abstract

This paper studies honest confidence sets for a transported complier average causal effect in a two-sample encouragement design. A source population supplies an instrumental-variables experiment with noncompliance, while a target population supplies the covariate law. Under the transported-IV model class, with source contrasts satisfying the stated measurability and integrability conditions, the transported Wald ratio equals the target complier contrast and lies in the compact range $[-1, 1]$. The paper characterizes the oracle minimax expected-length frontier for honest confidence sets over transported triangular arrays satisfying the stated overlap, sample-size, envelope, second-moment, and degrading-array conditions, with weight second moment κ_n and transported first-stage mean μ_n . The frontier is indexed by the effective strength

$$t_n = \frac{n\mu_n^2}{\kappa_n}.$$

For every fixed threshold $t_0 > 0$, oracle procedures that know the transport weight and source propensity have minimax expected length of order

$$\min\{1, t_0^{-1/2}\}.$$

A transported Anderson–Rubin score-inversion set attains this order. The unit-transport-weight least-favorable slice is embedded in the transported class, so its weak-ratio lower bound applies to the larger transported model. Up to constants depending on the sampling ratio, overlap, and noncoverage level, the same order holds conditionally on any admissible fixed covariate-shift geometry. In finite-cell designs, target cell frequencies and a collision-scale U-statistic yield feasible score-inversion procedures that learn the target weights from the target covariate sample and attain the same order under the stated cell-growth and overlap conditions.

1 Introduction

Encouragement experiments with imperfect take-up identify complier effects in the source population under the usual instrumental-variables restrictions [Bloom, 1984, Imbens and Angrist, 1994, Angrist et al., 1996]. Many applications also require a target-population interpretation: the source experiment is observed in one population, while the policy-relevant covariate distribution belongs to another. Transportability and generalizability methods address this covariate-shift problem through support and weighting conditions [Joseph Hotz et al., 2005, Cole and Stuart, 2010, Stuart et al., 2011, Tipton, 2013, Dahabreh et al., 2020b,a, Nguyen et al., 2017, Pearl and Bareinboim, 2011, Bareinboim and Pearl, 2013, 2016]. For transported noncompliance, the estimand is a ratio: a transported reduced form divided by a transported first stage. This ratio

structure makes weak identification central when the target complier share is small or when transport weights are dispersed.

This paper characterizes honest expected length for that transported complier ratio. The source sample contains (X, Z, D, Y) from an encouragement design, and the target sample contains covariates X . The target-to-source density ratio $w = dP_T/dP_S^X$ transports source conditional contrasts to the target covariate law. Under the transported-IV model class $\mathcal{P}_n$ in [definition 3](#), with source contrasts satisfying the stated measurability and integrability conditions, [theorem 1](#) shows that the transported reduced form equals the target complier-weighted outcome contrast, the transported first stage equals the target complier share, and their ratio equals the target complier effect θ_T . Bounded outcomes and binary treatment receipt place θ_T in $\Theta = [-1, 1]$, so expected length is measured on a fixed compact parameter space.

The main statistical object is the effective strength

$$t_n = \frac{n\mu_n^2}{\kappa_n},$$

where μ_n is the transported first-stage mean and $\kappa_n = \mathbb{E}_S[w(X)^2]$ is the Kish-type dispersion scale of the transport weights. This scalar combines two sources of difficulty. A small μ_n is the familiar weak-first-stage problem from IV ratio inference [[Anderson and Rubin, 1949](#), [Fieller, 1954](#), [Staiger and Stock, 1997](#), [Stock et al., 2002](#), [Moreira, 2003](#), [Andrews et al., 2006](#), [Mikusheva, 2010](#), [Andrews et al., 2019](#)]. A large κ_n records the variance inflation from transporting the source experiment to a different target covariate law. The frontier conditions allow $k_n \rightarrow \infty$, $k_n = o(\sqrt{n})$, and $\mu_n \rightarrow 0$, so the analysis covers triangular arrays in which covariate shift grows while the transported first stage weakens.

The oracle benchmark allows procedures to use the transport weight w and the source propensity e . [Theorem 2](#) proves that every oracle honest confidence-set sequence over $\mathcal{P}_n$ has limiting worst-case expected length at least a constant times

$$\min\{1, t_0^{-1/2}\}$$

over laws satisfying $t_n \geq t_0$. The same lower bound already holds on the uniform finite-cell submodel $\mathcal{N}_n$. [Theorem 3](#) gives the matching construction: invert the transported residual score

$$\frac{1}{n} \sum_{i=1}^n w(X_i) \left\{ \frac{Z_i}{e(X_i)} - \frac{1 - Z_i}{1 - e(X_i)} \right\} (Y_i - \vartheta D_i)$$

with a radius proportional to $\sqrt{\widehat{\kappa}_n/n}$. This Anderson–Rubin style inversion is honest over $\mathcal{P}_n$ and attains the same expected-length order for every fixed $t_0 > 0$.

The fixed-geometry results isolate the role of covariate shift. For an admissible deterministic geometry $\mathfrak{g} = (P_S^X, w, e)_{n \geq 1}$, [definition 16](#) holds the source covariate law, target weight, and propensity fixed while allowing the causal response law to vary. [Theorem 4](#) establishes the same two-sided order

$$V_{\mathfrak{g}}^*(t_0) \asymp \min\{1, t_0^{-1/2}\}$$

with constants uniform over admissible geometries. The lower bound uses the least-favorable family in [definition 19](#): the complier probability is proportional to the transport weight, so the transported first stage is calibrated to t_0 , and a local complier-outcome tilt moves the target ratio while the observed-data laws remain close in total variation as recorded in [definition 23](#).

[Theorem 5](#) specializes this comparison to $w = 1$, where $\kappa_n = 1$, $P_T = P_S^X$, and the effective strength reduces to $n\mu_n^2$.

The feasible results replace oracle-known weights by target-sample learning in finite-cell designs. In the uniform finite-cell class $\mathcal{N}_n$, the source covariate law assigns mass $1/k_n$ to each cell and the source propensity equals $1/2$. [Theorem 6](#) constructs a sample-only score-inversion set using target empirical cell frequencies, source cellwise reduced-form and first-stage contrasts, and a target-sample U-statistic for the collision scale. Under $N_n/n \rightarrow c > 0$ and $k_n = o(\sqrt{n})$, this feasible procedure is honest over $\mathcal{N}_n$ and attains the oracle frontier order. [Theorem 7](#) extends the construction to regular finite-cell designs with known source cell probabilities bounded between fixed multiples of $1/k_n$ and known cell-varying propensities satisfying overlap.

The contribution is therefore a matched minimax expected-length characterization for transported complier ratios under the stated transported-IV, overlap, positivity, support, envelope, dispersion, sample-size, and cell-growth conditions. The analysis connects transported noncompliance methods such as [Rudolph and Laan \[2016\]](#) and [Chen and Huang \[2025\]](#) with weak-identification-robust LATE and two-sample IV inference [[Choi et al., 2018](#), [Choi and Shen, 2019](#), [Ma, 2023](#), [Smucler et al., 2025](#)]. For the compact transported complier ratio under the stated transported-IV restrictions and growth conditions, the frontier is indexed by $n\mu_n^2/\kappa_n$, so first-stage weakness and transport-weight dispersion enter through a common scale.

The appendix gives the proof details and a verification note for the checked mathematical statements; see [section 6](#).

2 Setup, Transport, and Honest Risk

At row n , the two-population full-data potential-outcome world is governed by P_n^F on S , X , $D(0)$, $D(1)$, and $Y(0)$, $Y(1)$. The source population, $S = 1$, supplies an encouragement experiment with observed variables Z , D , and Y ; the target population, $S = 0$, supplies the covariate law to which the complier effect is transported. The measurable covariate space is $\mathcal{X}$. Potential outcomes are bounded because the expected-length problem is posed over a compact causal range, and treatment receipt is binary because compliance type is the margin through which the encouragement identifies the complier contrast.

Definition 1. [[synth_1](#)] (Inverse-propensity instrument score $H(O)$) *For a source observation $O = (X, Z, D, Y)$ and source propensity e , the inverse-propensity instrument score is the function H given by*

$$H(O) = \frac{Z}{e(X)} - \frac{1 - Z}{1 - e(X)}.$$

The score in [definition 1](#) is the source inverse-propensity contrast used to recover conditional reduced-form and first-stage contrasts under the source experiment. Its normalization keeps the assignment contrast centered conditionally on covariates when the source propensity $e(X)$ is bounded away from zero and one.

The full-data and sampling primitives are fixed by the following population and array conditions.

Assumption 1. [[ass:full-data-support](#)] (Full-data support) *The full-data law satisfies*

$$P_n^F \{S \in \{0, 1\}, D(0), D(1) \in \{0, 1\}, Y(0), Y(1) \in [0, 1]\} = 1.$$

This support restriction is specific to this analysis: it places the ratio parameter on a bounded outcome scale and makes the induced complier contrast live in a fixed compact interval.

Assumption 2. `[ass:population-presence]` (Population presence) *For each $s \in \{0, 1\}$,*

$$P_n^F(S = s) > 0.$$

Population presence is the standard two-population support condition, ensuring that the source and target conditional laws are well defined [Chen and Huang, 2025].

Assumption 3. `[ass:two-sample-array]` (Two-sample array) *For a transported triangular array on the measurable covariate space $\mathcal{X}$, let P_S denote the index- n observed source law induced on $O_i^S = (X_i, Z_i, D_i, Y_i)$, and let P_T denote the index- n target covariate law on X_j^T . The two sample factors satisfy:*

- *(Source factor.) For every n , P_S is a probability measure.*
- *(Target factor.) For every n , P_T is a probability measure.*
- *(Sample-size ratio.) $0 < c$ and $N_n/n \rightarrow c$.*

The observed two-sample triangular-array sampling world treats the source observations O_i^S and target covariates X_j^T as the two empirical inputs, with source law P_S , target covariate law P_T , target sample size N_n , and positive sample-size limit c . This is the standard independent two-sample triangular-array sampling condition used in transported noncompliance designs [Chen and Huang, 2025].

The next block gives the source instrumental-variables structure. These restrictions follow the usual LATE logic for an encouragement design, adapted to the transported two-population notation of the present problem [Imbens and Angrist, 1994, Angrist et al., 1996].

Assumption 4. `[ass:instrument-overlap]` (Instrument overlap) *The instrument-overlap constant satisfies $0 < \varepsilon < 1/2$, and the source propensity score satisfies*

$$\varepsilon \leq e(X) \leq 1 - \varepsilon$$

P_S^X -almost surely.

Instrument overlap is the standard instrument positivity condition, here indexed by the constant ε , and it keeps inverse-propensity contrasts uniformly controlled [Chen and Huang, 2025].

Assumption 5. `[ass:source-observation]` (Source observation law) *Conditional on $S = 1$, the observed source law satisfies:*

- *(Assignment propensity.) $P(Z = 1 \mid X, S = 1) = e(X)$.*
- *(Treatment receipt.) $D = D(Z)$.*
- *(Observed outcome.) $Y = Y(D)$.*

The source observation law is the standard assignment and consistency condition linking the observed source experiment to receipt and outcome potential outcomes [Chen and Huang, 2025].

Assumption 6. `[ass:iv-randomization]` (Instrument randomization) *The instrument satisfies*

$$Z \perp (Y(0), Y(1), D(0), D(1)) \mid (X, S = 1).$$

Instrument randomization is the standard conditional instrument exchangeability condition; within source covariate strata, the encouragement is as good as randomized for the relevant potential outcomes [Chen and Huang, 2025].

Assumption 7. `[ass:iv-exclusion]` (Instrument exclusion) *For every $z, s \in \{0, 1\}$,*

$$Y^Z(z) = Y(D(z))$$

P_n^F -almost surely conditional on $S = s$.

The exclusion restriction is the standard condition that the assignment potential outcome $Y^Z(z)$ operates through treatment receipt [Chen and Huang, 2025].

Assumption 8. `[ass:iv-monotonicity]` (Instrument monotonicity) *For every $s \in \{0, 1\}$,*

$$D(1) \geq D(0)$$

P_n^F -almost surely conditional on $S = s$.

Monotonicity is the standard no-defiers condition, allowing the receipt contrast to be interpreted as a complier share in each population state [Chen and Huang, 2025].

Transport enters through an outcome-contrast restriction, an integrated first-stage restriction, and covariate-support inclusion. The outcome condition is conditional in X , whereas the receipt condition is imposed at the target average; this distinction is the transport structure used by the identification result below.

Assumption 9. `[ass:outcome-transport]` (Outcome-contrast transport) *For P_T -almost every x ,*

$$\mathbb{E}_{P_n^F}[Y^Z(1) - Y^Z(0) \mid X = x, S = 0] = \mathbb{E}_{P_n^F}[Y^Z(1) - Y^Z(0) \mid X = x, S = 1].$$

Outcome-contrast transport is the standard conditional transport of the assignment-outcome contrast across the two population states [Chen and Huang, 2025].

Assumption 10. `[ass:receipt-transport]` (Receipt-contrast transport) *The target-population average of the conditional treatment-receipt contrast agrees across the two population states:*

$$\mathbb{E}_{X|S=0}[\mathbb{E}_{P_n^F}[D(1) - D(0) \mid X, S = 0]] = \mathbb{E}_{X|S=0}[\mathbb{E}_{P_n^F}[D(1) - D(0) \mid X, S = 1]].$$

Receipt-contrast transport equates the target-law average compliance margin across populations while allowing covariate-specific receipt contrasts to differ [Chen and Huang, 2025]. This condition is the denominator link that represents the target complier share by the source conditional first stage averaged over P_T .

Assumption 11. `[ass:target-complier-positivity]` (Target complier positivity) *The target population has positive complier mass:*

$$P_n^F(D(1) > D(0) \mid S = 0) > 0.$$

Target complier positivity is the standard positive target complier share condition, so the transported complier ratio is well defined [Chen and Huang, 2025].

Assumption 12. [ass:transport-domination] (Transport domination) *The target covariate law is absolutely continuous with respect to the source covariate marginal on $\mathcal{X}$:*

$$P_T \ll P_S^X.$$

Transport domination is the standard covariate-support inclusion condition for generalizing source contrasts to the target covariate distribution [Chen and Huang, 2025]. It yields the transport and overlap geometry through the source marginal P_S^X , the density-ratio weight $w(X) = dP_T/dP_S^X(X)$, and the source propensity.

The next definition collects the identification functionals built from this geometry: the source conditional contrasts, their transported means, and the effective-strength index.

Definition 2. [synth_6] (Transported contrasts and effective strength) *Let $w(X) = dP_T/dP_S^X(X)$ be a nonnegative version of the transport density ratio. The source conditional assignment-outcome and receipt contrasts are*

$$\Delta_Y(x) = \mathbb{E}_{P_n^F}[Y^Z(1) - Y^Z(0) \mid X = x, S = 1], \quad \Delta_D(x) = \mathbb{E}_{P_n^F}[D(1) - D(0) \mid X = x, S = 1].$$

Their transported means and the transport-weight dispersion are

$$\mu_{Y,n} = \mathbb{E}_{P_S^X}[w(X) \Delta_Y(X)], \quad \mu_n = \mathbb{E}_{P_S^X}[w(X) \Delta_D(X)], \quad \kappa_n = \mathbb{E}_{P_S^X}[w(X)^2],$$

and the effective identification strength at index n is $t_n = n\mu_n^2/\kappa_n$.

The remaining restrictions define the weak-first-stage geometry. They separate the size of the transport weights from the transported first stage, producing a single effective-strength index for honest expected length.

Assumption 13. [ass:weight-envelope] (Weight envelope) *The transport weight satisfies*

$$0 \leq w(X) \leq 2k_n$$

P_S -almost surely.

The weight envelope is specific to this analysis: the sequence k_n controls the largest admissible density-ratio values while permitting the amount of covariate shift to grow with n .

Assumption 14. [ass:weight-second-moment] (Weight second moment) *The second moment of the transport weight satisfies*

$$\kappa_n \leq k_n.$$

The second-moment bound is specific to this analysis and records the Kish-type dispersion scale $\kappa_n = \mathbb{E}_S[w(X)^2]$ that governs the variance of transported inverse-propensity averages.

Assumption 15. [ass:degrading-array] (Degrading array rates) *The array satisfies*

$$k_n \rightarrow \infty, \quad k_n = o(n^{1/2}), \quad \mu_n \rightarrow 0.$$

These rate restrictions are specific to this analysis: the envelope scale diverges at a sub-root- n rate, and the transported first-stage mean μ_n enters the weak-identification regime. Together with the dispersion scale, they define the effective-strength scalar $t_n = n\mu_n^2/\kappa_n$ used in the risk comparison.

Assumption 16. `[ass:finite-cell-source]` (Uniform finite cells) *The source covariate marginal P_S^X is supported on exactly k_n measurable cells, assigns mass $1/k_n$ to each cell, and satisfies*

$$e(X) = 1/2$$

P_S -almost surely.

The uniform finite-cell condition is specific to this analysis. It creates the sample-only cell-learning submodel used for feasible procedures, with equal source cell masses and a randomized source encouragement probability.

The full model and the finite-cell submodel collect these assumptions at each row of the triangular array.

Definition 3. `[def:model-class]` (Transported LATE model class $\mathcal{P}_n$) *The transported LATE model class at index n is*

$$\mathcal{P}_n = \{(P_n^F, e) : (P_n^F, e) \text{ satisfies the following conditions at index } n\}.$$

The required conditions are:

- *(Full-data support.)* [assumption 1](#).
- *(Population presence.)* [assumption 2](#).
- *(Two-sample array.)* [assumption 3](#).
- *(Instrument overlap.)* [assumption 4](#).
- *(Source observation.)* [assumption 5](#).
- *(Instrument randomization.)* [assumption 6](#).
- *(Instrument exclusion.)* [assumption 7](#).
- *(Instrument monotonicity.)* [assumption 8](#).
- *(Outcome transport.)* [assumption 9](#).
- *(Receipt transport.)* [assumption 10](#).
- *(Target complier positivity.)* [assumption 11](#).
- *(Transport domination.)* [assumption 12](#).
- *(Weight envelope.)* [assumption 13](#).
- *(Weight second moment.)* [assumption 14](#).
- *(Degrading array.)* [assumption 15](#).

The transported LATE model class $\mathcal{P}_n$ is the main domain for the oracle minimax statements. It combines the source IV restrictions with the transport and dispersion conditions that make the ratio target comparable across rows.

The finite-cell class is the transported model with one added row restriction:

$$\mathcal{N}_n = \{P \in \mathcal{P}_n : P \text{ satisfies } \text{assumption 16}\}.$$

Thus the IV, transport, overlap, envelope, second-moment, and degrading-array conditions are inherited from $\mathcal{P}_n$. The added content is the uniform source-cell geometry in [assumption 16](#); the anchored checklist form of $\mathcal{N}_n$ is deferred to [section 6](#).

Before defining risk, we record the oracle inputs and the two score radii used later. Oracle procedures receive the source sample, the target covariate sample, the transport weight, and the source propensity.

Definition 4. [\[synth_2\]](#) (Score-inversion critical radius L_α) *We define the score-inversion critical radius by $L_\alpha = \{8/(\alpha\varepsilon^2)\}^{1/2}$, where $\alpha \in (0, 1)$ is the noncoverage level and ε is the source instrument-overlap constant.*

The constant in [definition 4](#) combines the noncoverage level α with instrument overlap. It will determine the radius of the score-inversion set in the oracle construction.

The coverage requirements below are stated for the following causal estimand.

Definition 5. [\[synth_7\]](#) (Target complier effect θ_T) *The target complier effect is the ratio*

$$\theta_T = \frac{\mathbb{E}_{P_n^F}[\{Y(1) - Y(0)\} \mathbf{1}\{D(1) = 1, D(0) = 0\} \mid S = 0]}{P_n^F\{D(1) = 1, D(0) = 0 \mid S = 0\}},$$

the mean effect of treatment receipt among target-population compliers. Under target complier positivity the denominator is positive and $\theta_T = \mathbb{E}_{P_n^F}[Y(1) - Y(0) \mid D(1) = 1, D(0) = 0, S = 0]$; when the denominator is zero, the ratio is taken to be zero.

Definition 6. [\[def:oracle-honesty\]](#) (Oracle honest class $\mathfrak{C}^{\text{or}}$) *The oracle honest class is the class of oracle procedures $(C_n)_{n \geq 1}$ satisfying $0 < \alpha < 1$ and the following conditions. For every n , C_n maps each input*

$$\left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w, e \right), \quad w : \mathcal{X} \rightarrow \mathbb{R}, \quad w(x) \geq 0, \quad e : \mathcal{X} \rightarrow \mathbb{R},$$

to a subset of $\Theta = [-1, 1]$. For every n and every measurable such pair (w, e) , the sample-by-parameter graph

$$\left\{ \left(((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}), \vartheta \right) : \vartheta \in C_n \left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w, e \right) \right\}$$

is measurable. For every n , every $P \in \mathcal{P}_n$, and every pair w, w' of measurable nonnegative versions of dP_T/dP_S^X , the evaluations using the row- n propensity score agree for the two-sample law almost surely:

$$C_n \left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w, e \right) = C_n \left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w', e \right).$$

Finally,

$$\liminf_{n \rightarrow \infty} \begin{cases} \inf_{P \in \mathcal{P}_n} P(\theta_T \in C_n), & \mathcal{P}_n \neq \emptyset, \\ 1, & \mathcal{P}_n = \emptyset \end{cases} \geq 1 - \alpha,$$

where in the probability term C_n is evaluated at the canonical row- n density ratio dP_T/dP_S^X and the row- n propensity score.

The procedure and minimax-risk world treats each confidence set C_n as a random subset of the compact parameter space $\Theta = [-1, 1]$. Oracle honesty, written $\mathfrak{C}^{\text{or}}$, requires uniform asymptotic coverage over $\mathcal{P}_n$ while allowing the procedure to use the canonical nonnegative weight input w and the propensity.

Definition 7. `[def:frontier-risk]` (Frontier risk $R((C_n), t_0)$) *For a confidence-set sequence $(C_n)_{n \geq 1}$ and threshold t_0 , the limiting worst-case expected length above effective strength t_0 is*

$$R((C_n), t_0) = \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)].$$

The risk $R((C_n), t_0)$ measures limiting worst-case Lebesgue length $\lambda(C_n)$ over laws whose effective strength exceeds the positive threshold t_0 . This conditioning by t_n is the paper's way of comparing honest procedures at a fixed level of transported identification strength.

Definition 8. `[def:oracle-value]` (Oracle value $V^*(t_0)$) *For any fixed effective-strength threshold t_0 with $0 < t_0$, the oracle minimax expected-length value is*

$$V^*(t_0) = \inf_{(C_n) \in \mathfrak{C}^{\text{or}}} R((C_n), t_0).$$

The value $V^*(t_0)$ is the oracle benchmark over honest procedures with known transport weights and propensities.

Definition 9. `[def:finite-cell-oracle-value]` (Finite-cell oracle value $V_{\mathcal{N}}^*(t_0)$) *The finite-cell oracle minimax expected-length value at threshold t_0 is*

$$V_{\mathcal{N}}^*(t_0) = \inf_{(C_n) \in \mathfrak{C}^{\text{or}}} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)].$$

The finite-cell oracle value $V_{\mathcal{N}}^*(t_0)$ applies the same oracle procedure class to the uniform finite-cell submodel. It gives the comparison point for feasible cell-based weight learning.

Definition 10. `[def:finite-cell-feasible-honesty]` (Feasible finite-cell class $\mathfrak{C}^{\text{cell}}$) *The feasible finite-cell honest class is*

$$\mathfrak{C}^{\text{cell}} = \left\{ (C_n)_{n \geq 1} : C_n = C_n \left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n} \right) \subseteq \Theta, \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{N}_n} P(\theta_T \in C_n) \geq 1 - \alpha \right\}.$$

The feasible class $\mathfrak{C}^{\text{cell}}$ uses the source observations and target covariates as the procedure inputs in the uniform finite-cell design. Its coverage requirement remains uniform over $\mathcal{N}_n$.

The identification result connects the transported functionals of [definition 2](#) to the target complier effect of [definition 5](#): the inverse-propensity score recovers the source conditional contrasts Δ_Y and Δ_D from observed source moments, transport moves $\mu_{Y,n}$ and μ_n to target complier functionals, and the Wald ratio equals θ_T .

Theorem 1. `[prop:compact-causal-range]` (Compact causal range) *Fix n , and let P be a transported array with sample-size and envelope schedules N_n and k_n , constants c and ε , and model class $\mathcal{P}_n$ as in [definition 3](#). Suppose that:*

- **(Model membership.)** $P \in \mathcal{P}_n$ at the index n .
- **(Source contrasts.)** The assignment outcome contrast and the receipt contrast for the source population satisfy the usual measurability and integrability conditions under the source covariate marginal P_S^X .

Then Δ_Y and Δ_D represent the observed source outcome and receipt contrasts: for every measurable covariate set A ,

$$\int_{\{O: X \in A\}} H(O)Y dP_S(O) = \int_A \Delta_Y(x) dP_S^X(x), \quad \int_{\{O: X \in A\}} H(O)D dP_S(O) = \int_A \Delta_D(x) dP_S^X(x).$$

Moreover, the transported reduced-form outcome mean satisfies

$$\mu_{Y,n} = \mathbb{E}_{P_n^F}[\{Y(1) - Y(0)\} \mathbf{1}\{D(1) = 1, D(0) = 0\} \mid S = 0],$$

the transported first-stage mean satisfies

$$\mu_n = P_n^F\{D(1) = 1, D(0) = 0 \mid S = 0\},$$

and the Wald ratio equals the target complier-conditional contrast,

$$\frac{\mu_{Y,n}}{\mu_n} = \theta_T.$$

Finally, $\theta_T \in \Theta = [-1, 1]$.

Theorem 1 establishes the identification link used throughout the paper: the weighted source reduced form equals the target complier-weighted outcome contrast, the weighted source first stage equals the target complier share, and their ratio is the target complier-conditional contrast. The bounded outcome support then places the estimand in Θ , making expected length on a fixed compact parameter space the relevant risk criterion.

For later fixed-geometry arguments, we also record the notation for the centered least-favorable law and the two distance measures used to compare local alternatives.

Definition 11. [synth_5] (Centered fixed-geometry law $Q_{n,0}^g$) We write $Q_{n,0}^g$ for the $h = 0$ member of the fixed-geometry least-favorable family: for the fixed admissible geometry $\mathbf{g} = (P_S^X, w, e)_{n \geq 1}$, conditional on $X = x$ the compliance types $(D(0), D(1)) = (0, 1), (1, 1), (0, 0)$ have probabilities $p_n(x), (1 - p_n(x))/2, (1 - p_n(x))/2$, $Y(0) = 0$, $Y(1)$ is Bernoulli(1/2) for every compliance type, the same conditional full-data law is used in both populations, source encouragement is drawn with propensity $e(X)$, and the notation denotes the resulting joint observed-data law together with the fixed oracle inputs.

The centered law in [definition 11](#) is the baseline member of the fixed-geometry least-favorable family used in the lower-bound construction.

Definition 12. [synth_3] (Chi-square separation of the fixed-geometry submodel) For the fixed-geometry laws $Q_{n,h}^g$ and $Q_{n,0}^g$, we write $\chi^2(Q_{n,h}^g \| Q_{n,0}^g)$ for the chi-square divergence

$$\chi^2(Q_{n,h}^g \| Q_{n,0}^g) = \int \left(\frac{dQ_{n,h}^g}{dQ_{n,0}^g} - 1 \right)^2 dQ_{n,0}^g,$$

whenever $Q_{n,h}^g \ll Q_{n,0}^g$, with value $+\infty$ otherwise.

Definition 13. `[synth_4]` (Total variation separation of least-favorable laws) *We write $\|Q_{n,h}^{\mathfrak{g}} - Q_{n,0}^{\mathfrak{g}}\|_{\text{TV}}$ for the total variation distance between the index- n joint observed-data laws with fixed oracle inputs generated by the geometry $\mathfrak{g}$ at local complier-outcome tilt h and at the center $h = 0$, namely $\sup_A |Q_{n,h}^{\mathfrak{g}}(A) - Q_{n,0}^{\mathfrak{g}}(A)|$ over measurable events A in the corresponding source-sample, target-covariate-sample, and oracle-input space.*

The separations in [definitions 12](#) and [13](#) calibrate the statistical distance between fixed-geometry local alternatives. They provide the notation needed to express how lower-bound perturbations move the transported complier contrast while remaining close in observed-data law.

3 Oracle Frontier and Score Inversion

The minimax comparison now turns from identification to expected length. The relevant scalar is the effective strength $t_n = n\mu_n^2/\kappa_n$: the numerator records the transported first-stage signal and the denominator records the Kish dispersion of the transport weights. Weak-IV robust confidence sets are naturally built by inverting a reduced-form score restriction, following the same principle behind Anderson–Rubin and Fieller inference [[Anderson and Rubin, 1949](#), [Fieller, 1954](#), [Staiger and Stock, 1997](#), [Andrews et al., 2019](#)]. In the transported problem, the oracle score weights each source observation by the known density ratio and uses the known source propensity.

Definition 14. `[def:inversion-handle]` (Score-inversion set C_n) *For each $\vartheta \in \Theta$, define the source score*

$$w(X) \left\{ \frac{Z}{e(X)} - \frac{1-Z}{1-e(X)} \right\} (Y - \vartheta D).$$

Define the empirical second moment of the oracle weights by

$$\hat{\kappa}_n = \frac{1}{n} \sum_{i=1}^n w(X_i)^2.$$

With oracle-known w and e , the score-inversion confidence set is

$$C_n = \left\{ \vartheta \in \Theta : \left| \frac{1}{n} \sum_{i=1}^n w(X_i) \left\{ \frac{Z_i}{e(X_i)} - \frac{1-Z_i}{1-e(X_i)} \right\} (Y_i - \vartheta D_i) \right| \leq L_\alpha \sqrt{\frac{\hat{\kappa}_n}{n}} \right\}.$$

For oracle-known w and e , this score is a source-sample statistic.

The set in [definition 14](#) contains exactly those candidate ratios whose transported reduced-form residual is small on the scale $L_\alpha \sqrt{\hat{\kappa}_n}/n$. The radius uses the empirical second moment of the oracle weights, so covariate shift enters the score through the same dispersion term that appears in the effective-strength index. Because the construction is restricted to Θ , its length is measured on the compact causal range established in [theorem 1](#).

The first result gives the converse side of the oracle benchmark. It compares all oracle honest procedures in $\mathfrak{C}^{\text{or}}$ under the full transported class and under the uniform finite-cell submodel.

Theorem 2. `[thm:oracle-converse]` (Oracle frontier converse) *Fix $c > 0$, $\varepsilon \in (0, 1/2)$, and $\alpha \in (0, 1)$. There exist constants $c_0 > 0$ and $t_c > 0$, depending on c , ε , and α , such that the following holds.*

- **(Array size.)** The target sample sizes satisfy $N_n/n \rightarrow c$.
- **(Cell scale.)** The sequence k_n is positive, satisfies $k_n \rightarrow \infty$, and satisfies $k_n/\sqrt{n} \rightarrow 0$.
- **(Measurable finite carriers.)** For each n , the covariate space contains k_n distinct measurable singleton carrier cells.
- **(Transported-IV class.)** Every array in the transported model class $\mathcal{P}_n$ of [definition 3](#) satisfies [assumptions 1 to 15](#).
- **(Finite-cell submodel.)** Every array in the finite-cell class $\mathcal{N}_n$ of [definition 22](#) satisfies [assumption 16](#).

Then, for every $t_0 > 0$, the oracle values of [definitions 8 and 9](#) obey

$$c_0 \min\{1, t_0^{-1/2}\} \leq V^\star(t_0)$$

and

$$c_0 \min\{1, t_0^{-1/2}\} \leq V_{\mathcal{N}}^\star(t_0).$$

Moreover, for every $t_0 \in (0, t_c]$,

$$c_0 \leq V^\star(t_0).$$

The same constants c_0 and t_c apply uniformly over all decompositions of the effective strength t_0 into the transported first-stage mean μ_n and the weight dispersion κ_n .

[Theorem 2](#) establishes that honest expected length is bounded below at the order $\min\{1, t_0^{-1/2}\}$. The constants c_0 and t_c depend on the sampling ratio, overlap, and noncoverage level, while the final uniformity clause indexes the lower bound by the combined effective strength uniformly over decompositions into first-stage weakness and weight dispersion. The finite-cell inequality records the same frontier lower bound inside the uniform-cell submodel, which is the model used for feasible weight learning in the next section.

The matching construction is the score inversion from [definition 14](#). Its role is parallel to Anderson–Rubin confidence sets for weak instruments: each hypothesized ratio turns the problem into a reduced-form moment condition, and inversion of that moment yields an honest set even when the denominator is small [[Anderson and Rubin, 1949](#), [Dufour, 1997](#), [Stock et al., 2002](#), [Moreira, 2003](#), [Andrews et al., 2006](#), [Mikusheva, 2010](#), [Andrews et al., 2019](#)].

The attainment theorem uses an ambient nonemptiness hypothesis to ensure that the comparison domain contains an admissible geometry. The score-inversion sequence itself is global: when it is evaluated at a law in $\mathcal{P}_n$, it uses that law’s canonical density ratio and row- n propensity.

Theorem 3. [\[thm:oracle-score-inversion-attainment\]](#) (Oracle score attainment) *Fix sequences N_n and k_n , constants c, ε, α , the transported-IV model class $\mathcal{P}_n$ in [definition 3](#), and the score-inversion construction in [definition 14](#). Assume:*

- **(Sampling ratio.)** $c > 0$ and $N_n/n \rightarrow c$.
- **(Overlap and noncoverage.)** $0 < \varepsilon < 1/2$ and $0 < \alpha < 1$.
- **(Envelope scale.)** $k_n > 0$ for every n , $k_n \rightarrow \infty$, and $k_n/\sqrt{n} \rightarrow 0$.

- **(Nonempty admissible carrier.)** There is an admissible deterministic geometry $\mathfrak{g} \in \mathcal{G}$ satisfying [definition 15](#) at scale k_n and overlap constant ε .
- **(Uniform model restrictions.)** Every $P \in \mathcal{P}_n$ satisfies the two-sample condition in [assumption 3](#), the overlap condition in [assumption 4](#), the weight envelope in [assumption 13](#), the weight second-moment bound in [assumption 14](#), and the degrading-array condition in [assumption 15](#).

Let

$$L_\alpha = \left(\frac{8}{\alpha \varepsilon^2} \right)^{1/2}.$$

For each n , define

$$H_i = \frac{Z_i}{e(X_i)} - \frac{1 - Z_i}{1 - e(X_i)}, \quad \hat{A}_n = \frac{1}{n} \sum_{i=1}^n w(X_i) H_i Y_i, \quad \hat{B}_n = \frac{1}{n} \sum_{i=1}^n w(X_i) H_i D_i,$$

and

$$\hat{\kappa}_n = \frac{1}{n} \sum_{i=1}^n w(X_i)^2, \quad C_n = \left\{ \vartheta \in \Theta : \left| \hat{A}_n - \vartheta \hat{B}_n \right| \leq L_\alpha \sqrt{\frac{\hat{\kappa}_n}{n}} \right\}.$$

Then the score-inversion sequence (C_n) belongs to the oracle honest class $\mathfrak{C}^{\text{or}}$ of [definition 6](#). Moreover, with

$$C_0 = \max \left\{ 2, 4L_\alpha + \frac{8}{\varepsilon^2} \right\},$$

for every $t_0 > 0$, the frontier risk from [definition 7](#) satisfies

$$R((C_n), t_0) \leq C_0 \min\{1, t_0^{-1/2}\}.$$

Consequently the oracle value in [definition 8](#) satisfies the same attainable frontier order over fixed positive effective-strength thresholds.

[Theorem 3](#) gives the upper-bound side. The statistic $\hat{A}_n - \vartheta \hat{B}_n$ is the transported sample analogue of the reduced form for $Y - \vartheta D$, and the confidence set collects the values of ϑ whose score lies within the oracle radius. The constant C_0 is universal over the stated class once α and ε are fixed through L_α .

The carrier clause supplies the nonempty comparison domain used to state attainability. The transported score statistic in [theorem 3](#) uses the w and e supplied by each evaluated law.

Combining [theorems 2](#) and [3](#), the oracle minimax expected length is characterized, up to constants, by

$$\min\{1, t_0^{-1/2}\}.$$

For small effective strength, the compact parameter space yields a constant-length frontier. As t_0 grows, the length contracts at the inverse-square-root rate in the effective strength. The rate therefore treats a weak transported first stage and high transport dispersion symmetrically through $n\mu_n^2/\kappa_n$, which is the quantity carried forward to the feasible finite-cell procedures.

4 Fixed Geometry and Feasible Cell Weight Learning

The oracle comparison in [theorems 2](#) and [3](#) treats the transport weight and source propensity as inputs to the procedure. This section separates two questions that are useful for interpreting that benchmark. First, it holds the transport geometry fixed and records the local least-favorable family that delivers the same inverse-square-root dependence on effective strength. Second, it shows how the weight-dispersion scale can be learned from target covariates in finite-cell experiments while preserving the expected-length order.

A fixed geometry records the source covariate law, the target-to-source weight, and the source propensity as a deterministic triangular array. The following definition isolates the admissible arrays used throughout the section.

Definition 15. `[def:admissible-geometry-class]` (Admissible geometry class $\mathcal{G}$) *$\mathcal{G}$ is the class of deterministic arrays $\mathbf{g} = (P_S^X, w, e)_{n \geq 1}$ satisfying the following conditions:*

- *(Source law.)* P_S^X is a probability law on $\mathcal{X}$.
- *(Instrument overlap.)* $e : \mathcal{X} \rightarrow [\varepsilon, 1 - \varepsilon]$.
- *(Transport weights.)* $w : \mathcal{X} \rightarrow [0, 2k_n]$, $\mathbb{E}_S[w(X)] = 1$, and

$$\kappa_n = \mathbb{E}_S[w(X)^2] \leq k_n.$$

- *(Target law.)* For every measurable $A \subseteq \mathcal{X}$,

$$P_T(A) = \mathbb{E}_S[w(X)\mathbf{1}\{X \in A\}].$$

The class $\mathcal{G}$ fixes the $\mathbf{g}$ component of the transported experiment while retaining the same overlap, envelope, normalization, and second-moment controls used in the oracle frontier. The target law is generated from the source covariate law by the density ratio, so the geometry keeps the covariate-shift component explicit and leaves the causal response law to vary within the model.

Definition 16. `[def:fixed-geometry-slice]` (Fixed-geometry slice $\mathcal{P}_n(\mathbf{g})$) *For $\mathbf{g} \in \mathcal{G}$,*

$$\mathcal{P}_n(\mathbf{g}) = \{P \in \mathcal{P}_n : (P_S^X, w, e) \text{ equals the index-} n \text{ component of } \mathbf{g}\}.$$

The slice $\mathcal{P}_n(\mathbf{g})$ is the subproblem in which the causal and outcome laws vary subject to the transported LATE restrictions, while the source covariate marginal, target density ratio, and propensity are held fixed.

Definition 17. `[def:fixed-geometry-honesty]` (Fixed-geometry oracle honesty $\mathfrak{C}^{\text{or}}(\mathbf{g})$) *For $\mathbf{g} \in \mathcal{G}$ as in [definition 15](#), $\mathfrak{C}^{\text{or}}(\mathbf{g})$ is the class of sequences $(C_n)_{n \geq 1}$ such that*

$$C_n = C_n\left((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w, e\right) \subseteq \Theta$$

and

$$\liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}_n(\mathbf{g})} P(\theta_T \in C_n) \geq 1 - \alpha.$$

The class $\mathfrak{C}^{\text{or}}(\mathfrak{g})$ imposes honesty over the fixed-geometry slice. Procedures may use the fixed oracle inputs, and the coverage requirement remains uniform over the causal response laws compatible with that geometry.

Definition 18. [def:fixed-geometry-value] (Fixed-geometry value $V_{\mathfrak{g}}^*(t_0)$) For $\mathfrak{g} \in \mathcal{G}$ as in [definition 15](#), the fixed-geometry minimax expected-length value at threshold t_0 is

$$V_{\mathfrak{g}}^*(t_0) = \inf_{(C_n) \in \mathfrak{C}^{\text{or}}(\mathfrak{g})} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n(\mathfrak{g}): t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)].$$

The value $V_{\mathfrak{g}}^*(t_0)$ measures the same expected-length criterion as [definition 8](#), now conditional on a fixed admissible geometry. This formulation makes clear that the effective-strength comparison is a statement about the response-law variation remaining after the covariate-shift array has been fixed.

The lower-bound construction varies a single complier-outcome tilt while preserving the fixed geometry. It uses the same target-first-stage calibration as the oracle converse, with a local radius chosen so that the observed-data laws remain close in total variation.

Definition 19. [def:least-favorable-witness] (Least-favorable witness $Q_{n,h}^{\mathfrak{g}}$) Fix $\mathfrak{g} = (P_S^X, w, e)_{n \geq 1} \in \mathcal{G}$ and $t_0 > 0$. Set

$$\mu_n = \left(\frac{t_0 \kappa_n}{n} \right)^{1/2}, \quad p_n(x) = \frac{\mu_n w(x)}{\kappa_n}, \quad \rho = \frac{1 - \alpha}{8}, \quad H(t_0) = \min\{1/4, \rho t_0^{-1/2}\}.$$

For each $|h| \leq H(t_0)$, conditional on $X = x$, assign compliance types by

$$\Pr\{(D(0), D(1)) = (0, 1) \mid X = x\} = p_n(x),$$

$$\Pr\{(D(0), D(1)) = (1, 1) \mid X = x\} = \Pr\{(D(0), D(1)) = (0, 0) \mid X = x\} = \frac{1 - p_n(x)}{2}.$$

Set $Y(0) = 0$. For compliers, let $Y(1)$ be Bernoulli with mean $1/2 + h$; for always-takers and never-takers, let $Y(1)$ be Bernoulli with mean $1/2$. Use this same conditional full-data law in both populations, draw source encouragement with propensity $e(X)$, and write $Q_{n,h}^{\mathfrak{g}}$ for the resulting joint observed-data law together with the fixed oracle inputs.

In [definition 19](#), the complier probability $p_n(x)$ is proportional to the transport weight, so the transported first stage is exactly calibrated to the threshold t_0 . The parameter ρ fixes the Le Cam separation scale, and the tilt h moves the target complier contrast while preserving the same source and target covariate geometry.

The notation for the centered law and the two distances is fixed in [definitions 11 to 13](#). The fixed-geometry lower bound also uses the separation calculation recorded in [definition 23](#): a tilt of order $t_0^{-1/2}$ moves the transported complier contrast while the observed-data laws remain close in chi-square and total variation.

The next theorem gives the fixed-geometry frontier for arbitrary admissible deterministic geometries. It combines the lower-bound calibration in [definitions 19 and 23](#) with the oracle score construction from [theorem 3](#).

Theorem 4. [thm:fixed-geometry-frontier] (Fixed-geometry frontier) Let $c > 0$, $0 < \varepsilon < 1/2$, and $0 < \alpha < 1$. Assume the following conditions hold:

- **(Target-sample ratio.)** The target sample sizes satisfy $N_n/n \rightarrow c$.
- **(Envelope scale.)** The sequence k_n is positive for every n , satisfies $k_n \rightarrow \infty$, and obeys $k_n/\sqrt{n} \rightarrow 0$.
- **(Geometry.)** The deterministic geometry array $\mathbf{g}$ satisfies [definition 15](#) with scale k_n and overlap constant ε .
- **(Fixed-geometry slices.)** For every n and every transported array whose n th law lies in the fixed-geometry slice $\mathcal{P}_n(\mathbf{g})$ of [definition 16](#), the conditions in [assumptions 3, 4 and 13 to 15](#) hold.

For every $t_0 > 0$, define

$$L_\alpha = \left(\frac{8}{\alpha \varepsilon^2} \right)^{1/2}, \quad c_\alpha = \frac{3(1-\alpha)^2}{16}, \quad C_\alpha = \max \left\{ 2, 4L_\alpha + \frac{8}{\varepsilon^2} \right\}.$$

Then the fixed-geometry minimax value of [definition 18](#) satisfies

$$c_\alpha \min\{1, t_0^{-1/2}\} \leq V_{\mathbf{g}}^*(t_0) \leq C_\alpha \min\{1, t_0^{-1/2}\}.$$

The constants c_α and C_α are uniform over admissible deterministic geometries $\mathbf{g}$.

[Theorem 4](#) establishes that fixing the covariate-shift geometry leaves the same expected-length order as the full oracle comparison: $\min\{1, t_0^{-1/2}\}$. The constants are uniform over the admissible geometry class, so the effective-strength index absorbs both first-stage weakness and weight dispersion across the fixed-geometry slices.

A useful specialization arises when the target and source covariate laws coincide. In that case the dispersion term collapses to one, and the transported effective strength reduces to the familiar sample-size-scaled first-stage square.

Theorem 5. [prop:no-shift-reduction] (Single-population frontier reduction) *On a measurable covariate space $\mathcal{X}$, let $\mathbf{g}$ be a deterministic geometry array satisfying [definition 15](#) with scale k_n and overlap constant ε . Suppose that*

- **(Sampling ratio.)** $c > 0$ and $N_n/n \rightarrow c$.
- **(Overlap and noncoverage.)** $0 < \varepsilon < 1/2$ and $0 < \alpha < 1$.
- **(Cell scale.)** $k_n > 0$ for every n , $k_n \rightarrow \infty$, and $k_n/\sqrt{n} \rightarrow 0$.
- **(Unit transport weight.)** For every n , $w_n(X) = 1$ $P_{S,n}^X$ -almost surely under $\mathbf{g}$.

Then, for every n ,

$$\kappa_n = 1 \quad \text{and} \quad P_{T,n} = P_{S,n}^X.$$

Moreover, for every transported array P and every n , if $P \in \mathcal{P}_n(\mathbf{g})$ in the sense of [definition 16](#), then its effective strength satisfies

$$t_n = n\mu_n^2.$$

Finally, for every $t_0 > 0$, let

$$L_\alpha = \sqrt{\frac{8}{\alpha \varepsilon^2}}, \quad c_\alpha = \frac{3(1-\alpha)^2}{16}, \quad C_\alpha = \max \left\{ 2, 4L_\alpha + \frac{8}{\varepsilon^2} \right\}.$$

The fixed-geometry minimax value $V_g^*(t_0)$ from [definition 18](#) obeys

$$c_\alpha \min\{1, t_0^{-1/2}\} \leq V_g^*(t_0) \leq C_\alpha \min\{1, t_0^{-1/2}\}.$$

[Theorem 5](#) records the unit-weight specialization of [theorem 4](#). With unit transport weight, the target law equals the source covariate law, $\kappa_n = 1$, and the rate is indexed by $n\mu_n^2$. The same compact-parameter expected-length order therefore recovers the usual weak-first-stage scaling within the transported notation.

We now turn from oracle-known weights to sample-only finite-cell score inversion. In the uniform finite-cell model of [definition 22](#), the target sample identifies the target cell frequencies, while the source sample estimates the cellwise outcome and receipt contrasts. This is the finite-cell analogue of inverse-probability weighting in semiparametric transport and missing-data problems [[Hirano et al., 2003](#), [Robins et al., 1994](#), [van der Laan and Rubin, 2006](#), [Chernozhukov et al., 2018](#)].

Theorem 6. `[thm:finite-cell-unknown-weight-attainment]` (Finite-cell sample-only score-inversion attainment) *Let N_n and k_n be positive integer sequences, let $c, \varepsilon, \alpha \in \mathbb{R}$, and suppose the following conditions hold.*

- **(Sampling ratio.)** $c > 0$ and $N_n/n \rightarrow c$.
- **(Overlap and coverage levels.)** $0 < \varepsilon < 1/2$ and $0 < \alpha < 1$.
- **(Cell growth.)** $k_n > 0$ for every n , $k_n \rightarrow \infty$, and $k_n/\sqrt{n} \rightarrow 0$.
- **(Ambient carrier.)** For every n , the measurable covariate space contains k_n distinct measurable singleton cells.
- **(Finite-cell rows.)** For every row n and every transported array $P \in \mathcal{N}_n$, the row satisfies the two-sample condition in [assumption 3](#), the uniform finite-cell source condition in [assumption 16](#), and the degrading-array condition in [assumption 15](#), with $\mathcal{N}_n$ as in [definition 22](#).

Put

$$B_c = 32(1 + c^{-1}), \quad L_{\alpha,c} = \left(\frac{2B_c}{\alpha}\right)^{1/2}, \quad C_0 = \max\{2, 4\sqrt{2}L_{\alpha,c} + 8B_c\}.$$

Then $C_0 > 0$, and there exists a sample-only finite-cell confidence-set sequence $C = (C_n)$ in the feasible class $\mathfrak{C}^{\text{cell}}$ of [definition 10](#) such that, for every n and every $P \in \mathcal{N}_n$, C_n agrees P -almost surely with the finite-cell score-inversion rule using critical constant $L_{\alpha,c}$. Equivalently, on the finite-cell experiment, this rule is given by

$$H_i = 2(2Z_i - 1), \quad \hat{p}_{x,n} = \frac{1}{N_n} \sum_{j=1}^{N_n} \mathbf{1}\{X_j^T = x\},$$

and, for $G \in \{Y, D\}$,

$$\hat{m}_{G,n}(x) = \frac{k_n}{n} \sum_{i=1}^n \mathbf{1}\{X_i = x\} H_i G_i, \quad \widehat{M}_{G,n} = \sum_{x=1}^{k_n} \hat{p}_{x,n} \hat{m}_{G,n}(x).$$

For $N_n \geq 2$, define

$$\hat{\kappa}_{U,n} = \frac{k_n}{N_n(N_n - 1)} \sum_{j \neq \ell} \mathbf{1}\{X_j^T = X_\ell^T\}, \quad \hat{K}_n = 1 + \hat{\kappa}_{U,n},$$

and set

$$C_n = \left\{ \vartheta \in \Theta : |\widehat{M}_{Y,n} - \vartheta \widehat{M}_{D,n}| \leq L_{\alpha,c} \sqrt{\hat{K}_n/n} \right\},$$

with $C_n = \Theta$ at indices for which $N_n < 2$. For every $t_0 > 0$,

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)] \leq C_0 \min\{1, t_0^{-1/2}\}.$$

Moreover, for every $t_0 > 0$, the feasible finite-cell honest minimax risk satisfies

$$\frac{3(1-\alpha)^2}{16} \min\{1, t_0^{-1/2}\} \leq \inf_{\tilde{C} \in \mathfrak{C}^{\text{cell}}} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(\tilde{C}_n)].$$

Theorem 6 constructs a feasible confidence set from the source observations and target covariates. The empirical target frequencies $\hat{p}_{x,n}$ estimate the transported mixing weights, the cellwise contrasts $\hat{m}_{G,n}(x)$ estimate the source reduced-form and first-stage contrasts, and $\hat{\kappa}_{U,n}$ is a target-sample U-statistic for the weight-dispersion scale. The resulting inversion set has the same $\min\{1, t_0^{-1/2}\}$ expected-length order as the oracle benchmark on the uniform finite-cell class, with the lower bound showing that this order is minimax sharp within $\mathfrak{C}^{\text{cell}}$.

For the uniform-cell benchmark, the inputs are the source records (X, Z, D, Y) , the target covariates X^T , the known cell partition, and the known propensity $1/2$. Tabulate target frequencies, compute $H_i = 2(2Z_i - 1)$ and the source cell contrasts, aggregate the two transported moments, estimate dispersion with the target collision statistic, and invert the displayed affine score over Θ . The regular-cell version follows the same steps while taking the known $q_{x,n}$ and $e_{x,n}$ as additional inputs. These constructions are minimax attainability benchmarks under the stated growth conditions; their declared inputs fix the cell structure and propensities used by the rules.

The uniform-cell construction uses known equal source cell masses and the propensity $e(X) = 1/2$. A regular finite-cell variant allows unequal but controlled source cell probabilities and a known cell-varying propensity. The source-cell probabilities $q_{x,n}$ are bounded between constants c_- and c_+ times $1/k_n$, which preserves the same dispersion order while accommodating regular discrete designs.

Definition 20. [def:regular-finite-cell-class] (Regular finite-cell class $\mathcal{N}_n^{\text{reg}}$) For fixed constants $0 < c_- \leq 1 \leq c_+ < \infty$, define

$$\mathcal{N}_n^{\text{reg}} = \left\{ P \in \mathcal{P}_n : \begin{array}{l} P_S^X \text{ is supported on exactly } k_n \text{ measurable cells,} \\ q_{x,n} = P_S^X(X \text{ lies in cell } x), \\ c_-/k_n \leq q_{x,n} \leq c_+/k_n \text{ for every } x \end{array} \right\}.$$

The regular class $\mathcal{N}_n^{\text{reg}}$ keeps the finite support structure while allowing cell masses to vary within fixed multiplicative bounds. This is the discrete counterpart of overlap and bounded-density conditions commonly used in transport and weighting analyses [Chen and Huang, 2025, Rudolph and Laan, 2016, Zeng et al., 2024].

Definition 21. [def:regular-cell-feasible-honesty] (Regular-cell feasible class $\mathfrak{C}^{\text{reg}}$) For a noncoverage level $0 < \alpha < 1$, define $\mathfrak{C}^{\text{reg}}$ as the class of regular-cell procedures $(C_n)_{n \geq 1}$ with the following properties. For each n , C_n is a Θ -valued random set whose inputs are $(O_i^S)_{i=1}^n$, $(X_j^T)_{j=1}^{N_n}$, the realized regular-cell support, and the known cell-indexed vectors $(q_{x,n})_{x=1}^{k_n}$ and $(e_{x,n})_{x=1}^{k_n}$. For every realized regular-cell support and every pair of real k_n -vectors $(q_{x,n})_{x=1}^{k_n}$ and $(e_{x,n})_{x=1}^{k_n}$, the sample-by- θ graph of C_n , restricted to source and target samples on that support, is measurable. These procedures satisfy

$$1 - \alpha \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{N}_n^{\text{reg}}} P(\theta_T \in C_n),$$

with the rowwise infimum evaluated as 1 when $\mathcal{N}_n^{\text{reg}}$ is empty.

The feasible class $\mathfrak{C}^{\text{reg}}$ formalizes procedures that know the regular source-cell probabilities and the cell-indexed propensity, but learn the target cell frequencies from the target sample. The measurability clause ensures that the inversion rule is a well-defined random set on each realized finite carrier.

Theorem 7. [thm:regular-cell-unknown-weight-attainment] (Regular-cell weight attainment) Fix $c, \varepsilon, \alpha, c_-, c_+ \in \mathbb{R}$ and sequences N_n and k_n . Suppose that

- **(Sample-size ratio.)** $0 < c$ and $N_n/n \rightarrow c$.
- **(Overlap and coverage levels.)** $0 < \varepsilon < 1/2$ and $0 < \alpha < 1$.
- **(Regular cell constants.)** $0 < c_- \leq 1$ and $1 \leq c_+$.
- **(Cell growth.)** $k_n > 0$ for every n , $k_n \rightarrow \infty$, and $k_n/\sqrt{n} \rightarrow 0$.
- **(Measurable carrier.)** For every n , the covariate carrier contains k_n injected measurable singleton cells.
- **(Model restrictions.)** Every array $P \in \mathcal{N}_n^{\text{reg}}$ from [definition 20](#) with constants c, ε, c_-, c_+ satisfies [assumption 3](#), [assumption 4](#), and [assumption 15](#).

On $\mathcal{N}_n^{\text{reg}}$, let $q_{x,n} = P_S^X(X = x)$ and define

$$H_i = \frac{Z_i}{e(X_i)} - \frac{1 - Z_i}{1 - e(X_i)}.$$

For $G \in \{Y, D\}$, set

$$\hat{p}_{x,n} = \frac{1}{N_n} \sum_{j=1}^{N_n} \mathbf{1}\{X_j^T = x\}, \quad \hat{m}_{G,n}(x) = \frac{1}{nq_{x,n}} \sum_{i=1}^n \mathbf{1}\{X_i = x\} H_i G_i,$$

and

$$\widehat{M}_{G,n} = \sum_{x=1}^{k_n} \hat{p}_{x,n} \hat{m}_{G,n}(x).$$

For $N_n \geq 2$, set

$$\hat{\kappa}_{U,n} = \frac{1}{N_n(N_n - 1)} \sum_{j \neq \ell} \frac{\mathbf{1}\{X_j^T = X_\ell^T\}}{q_{X_j^T,n}}, \quad \hat{K}_n = 1 + \hat{\kappa}_{U,n},$$

and put $C_n^{\text{reg}} = \Theta$ when $N_n < 2$. Let

$$B_{\varepsilon,c} = 8(\varepsilon^{-2} + c^{-1}).$$

For every fixed $L \geq \{2B_{\varepsilon,c}/\alpha\}^{1/2}$, the sequence

$$C_n^{\text{reg}} = \left\{ \vartheta \in \Theta : |\widehat{M}_{Y,n} - \vartheta \widehat{M}_{D,n}| \leq L \sqrt{\widehat{K}_n/n} \right\}$$

for $N_n \geq 2$, together with the preceding convention for $N_n < 2$, belongs to $\mathfrak{C}^{\text{reg}}$ from [definition 21](#) and is constructed from the known regular source-cell probabilities, the known propensity, and the source and target samples. Moreover, for every fixed $t_0 > 0$,

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n^{\text{reg}}: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n^{\text{reg}})] \leq \max\{2, 4\sqrt{2}L + 8B_{\varepsilon,c}\} \min\{1, t_0^{-1/2}\}.$$

The corresponding minimax risk over $\mathfrak{C}^{\text{reg}}$ is bounded below by

$$\frac{3(1-\alpha)^2}{16} \min\{1, t_0^{-1/2}\}.$$

Thus, with known regular source-cell probabilities and a known cell-varying propensity satisfying the stated overlap, the feasible regular-cell construction attains the oracle frontier order on $\mathcal{N}_n^{\text{reg}}$.

[Theorem 7](#) extends finite-cell score inversion to regular source-cell masses and known cell-varying propensities. The statistic uses inverse-propensity source contrasts, target empirical cell frequencies, and a regular-cell U-statistic dispersion proxy. Under the stated sampling, overlap, and cell-growth conditions, the construction is honest over $\mathfrak{C}^{\text{reg}}$ and attains the same expected-length order as the fixed-geometry and uniform finite-cell procedures.

Together, [theorems 4, 6 and 7](#) show how the oracle frontier carries over from fixed weights to sample-learned finite-cell weights. The common rate $\min\{1, t_0^{-1/2}\}$ is indexed by the effective strength $n\mu_n^2/\kappa_n$, with κ_n interpreted either as an oracle dispersion measure or as a target-sample-estimated finite-cell collision scale.

5 Related Literature and Discussion

The paper belongs first to the LATE and principal-stratification literature. The source experiment uses the encouragement-design logic of [Bloom \[1984\]](#), [Imbens and Angrist \[1994\]](#), and [Angrist et al. \[1996\]](#): conditional randomization, exclusion, and monotonicity turn the source reduced form and first stage into a complier contrast. Principal stratification supplies the population-level interpretation of compliance types and target complier effects [[Balke and Pearl, 1997](#), [Imbens and Rubin, 1997](#), [Frangakis and Rubin, 2002](#), [Imbens and Rosenbaum, 2005](#), [Heckman and Vytlacil, 2005](#)]. In the present setting, [theorem 1](#) identifies the transported reduced form and transported first stage through weighted source contrasts and places the target complier contrast inside the compact parameter space $\Theta = [-1, 1]$.

The weak-identification component follows the robust ratio-inference tradition. Anderson–Rubin and Fieller procedures invert reduced-form score restrictions, giving ratio inference through the score restriction itself [[Anderson and Rubin, 1949](#), [Fieller, 1954](#)]. Subsequent work characterizes the consequences of weak instruments for distributional approximation, confidence

sets, and expected length [Gleser and Hwang, 1987, Dufour, 1997, Bound et al., 1995, Staiger and Stock, 1997, Stock et al., 2002, Moreira, 2003, Andrews et al., 2006, Mikusheva, 2010, Andrews et al., 2019, Bertanha and Moreira, 2020, Ma, 2023, Smucler et al., 2025]. The oracle score construction in [definition 14](#) is the transported analogue of this logic: it tests candidate values by centering the weighted reduced form after subtracting the candidate times the weighted first stage. [Theorems 2 and 3](#) then characterize the minimax expected-length order for the transported ratio as $\min\{1, t_0^{-1/2}\}$.

Transportability and generalizability enter through the target-to-source covariate weighting structure. Work on transporting and generalizing treatment effects emphasizes support, covariate reweighting, and target-population interpretation [Joseph Hotz et al., 2005, Pearl and Bareinboim, 2011, Bareinboim and Pearl, 2013, 2016, Cole and Stuart, 2010, Stuart et al., 2011, Tipton, 2013, Dahabreh et al., 2020b,a, Nguyen et al., 2017]. Here the transported-IV restrictions in [definition 3](#) combine those support and contrast conditions with an IV compliance structure. The effective-strength scalar $t_n = n\mu_n^2/\kappa_n$ summarizes the transported first-stage mean μ_n and Kish dispersion κ_n , so the expected-length comparison uses a single scale for weak compliance and covariate reweighting.

The feasible procedures connect to semiparametric weighting and modern transported noncompliance methods. Inverse-probability weighting, augmented estimation, and sample-splitting ideas motivate estimating transported nuisance components while retaining valid target-population interpretation [Hirano et al., 2003, Robins et al., 1994, van der Laan and Rubin, 2006, Chernozhukov et al., 2018].

Transported noncompliance. [Chen and Huang \[2025\]](#), cited as an arXiv working paper, identify and estimate transported CACE in a two-sample generalizability design with source noncompliance and target population information; their inferential object is transported-CACE identification and estimation. [Rudolph and Laan \[2016\]](#), a journal article, develop robust estimators for encouragement-design effects transported across sites. [Ross et al. \[2026\]](#), also a journal article, study trial transport when participation changes adherence, with transported trial effects as the inferential target. The present paper adds the honest minimax expected-length frontier for a compact transported-CACE ratio under a triangular-array weak first stage and Kish weight dispersion, together with finite-cell procedures that attain the oracle order.

Weak-IV and two-sample IV inference. [Ma \[2023\]](#), cited as an arXiv working paper, gives identification-robust LATE inference with high-dimensional covariates. [Smucler et al. \[2025\]](#), also cited as an arXiv working paper, studies the properties of LATE confidence sets obtained by score-test inversion. [Choi et al. \[2018\]](#) and [Choi and Shen \[2019\]](#) instead give weak-instrument-robust inference for linear IV coefficients assembled from two samples. The present paper’s source experiment and target covariate sample define a covariate-transport problem, and its positive scope is an honest frontier indexed by $n\mu_n^2/\kappa_n$ with finite-cell target-weight learning.

Relative to transported-ATE efficiency and minimax benchmarks, the parameter here is a compact transported complier ratio whose denominator may shrink along the triangular array. Under the stated transported-IV, bounded-outcome, overlap, and dispersion conditions, the frontier is indexed by the ratio strength $n\mu_n^2/\kappa_n$, whereas regular transported-ATE benchmarks use root- n target-mean scales.

Relative to classical weak-IV and two-sample IV inference, the present setting adds target-to-source covariate weighting and finite-cell target-law learning. Under the stated finite-cell and

regular-cell conditions, the expected-length order records first-stage weakness and the dispersion induced by transporting the covariate distribution through the single scale $n\mu_n^2/\kappa_n$.

Taken together, the results give a finite expected-length characterization for transported complier ratios in the stated transported-IV model classes, with the stated overlap, transport, positivity, envelope, dispersion, sample-size, and cell-growth conditions. The compact causal range in [theorem 1](#) keeps length measured on Θ , while [theorem 4](#) shows that the fixed-geometry minimax value is indexed by the effective-strength threshold t_0 . This synthesis places transported noncompliance on the same weak-ratio inference scale as classical weak-IV analysis, with κ_n recording the additional variation induced by moving from the source covariate law to the target covariate law.

6 Appendix: Proofs, Auxiliary Arguments, and Verification Note

The appendix gives the mathematical arguments in dependency order. It begins with the compact causal-range calculation, then proves the fixed-geometry frontier and its unit-weight specialization, then gives the oracle lower and upper bounds, and finally treats the regular and uniform finite-cell constructions. Throughout, confidence sets take values in the compact parameter space Θ , and expected length is the Lebesgue length criterion from [definition 7](#).

The finite-cell submodel is $\mathcal{P}_n$ plus the uniform finite-cell source condition in [assumption 16](#). For reference, the anchored formal checklist is:

Definition 22. `[def:finite-cell-class]` (Finite-cell class $\mathcal{N}_n$) *The finite-cell submodel at index n is*

$$\mathcal{N}_n = \{P \in \mathcal{P}_n : P \text{ satisfies the following conditions at index } n\}.$$

The required conditions are:

- *(Full-data support.)* [assumption 1](#).
- *(Population presence.)* [assumption 2](#).
- *(Two-sample array.)* [assumption 3](#).
- *(Instrument overlap.)* [assumption 4](#).
- *(Source observation.)* [assumption 5](#).
- *(Instrument randomization.)* [assumption 6](#).
- *(Instrument exclusion.)* [assumption 7](#).
- *(Instrument monotonicity.)* [assumption 8](#).
- *(Outcome transport.)* [assumption 9](#).
- *(Receipt transport.)* [assumption 10](#).
- *(Target complier positivity.)* [assumption 11](#).
- *(Transport domination.)* [assumption 12](#).

- (*Weight envelope.*) [assumption 13](#).
- (*Weight second moment.*) [assumption 14](#).
- (*Degrading array.*) [assumption 15](#).
- (*Finite-cell source.*) [assumption 16](#).

The fixed-geometry lower bounds use the centered law and distance notation in [definitions 11 to 13](#). The following anchored statement records the separation calculation for the least-favorable laws generated by [definition 19](#).

Definition 23. `[def:geometry-handle]` (Geometry handle $Q_{n,h}^g$) *Fix an admissible deterministic array $\mathbf{g} = (P_S^X, w, e)_{n \geq 1}$. For $t_0 > 0$, set*

$$\mu_n = \left(\frac{t_0 \kappa_n}{n} \right)^{1/2}, \quad p_n(x) = \frac{\mu_n w(x)}{\kappa_n}.$$

For $0 < \alpha < 1$, let $\rho = (1 - \alpha)/8$. As the complier outcome mean varies over $h \in \mathbb{R}$ with

$$|h| \leq \min\{1/4, \rho t_0^{-1/2}\},$$

using the chi-square divergence and total variation distance defined in [definitions 12 and 13](#), the corresponding least-favorable laws satisfy

$$1 + \chi^2(Q_{n,h}^g \| Q_{n,0}^g) \leq \exp(8t_0 h^2), \quad \|Q_{n,h}^g - Q_{n,0}^g\|_{\text{TV}} \leq 2\rho.$$

The proof of [theorem 1](#) expands the inverse-propensity score in [definition 1](#) conditional on X . Conditional instrument randomization and the source observation law turn the source score moments into the conditional assignment contrasts $\Delta_Y(x)$ and $\Delta_D(x)$. Exclusion and monotonicity express those contrasts through complier potential outcomes, while [assumptions 9, 10 and 12](#) move the reduced form and first stage to the target covariate law. The bounded outcome support in [assumption 1](#) gives the compact range $[-1, 1]$, so the transported Wald ratio is interpreted as the target complier-conditional contrast in [theorem 1](#).

The lower-bound arguments for [theorems 2 and 4](#) use the least-favorable family in [definition 19](#). That construction sets the complier probability proportional to the transport weight, which calibrates the transported first stage to the selected effective-strength threshold. Varying the complier outcome tilt changes the target complier contrast by a controlled amount, while [definition 23](#) bounds the chi-square and total-variation separations between the tilted law and the centered law. Le Cam’s testing argument then converts closeness of the observed-data laws into a lower bound on the expected length of any honest confidence set. The single-population specialization in [theorem 5](#) follows by setting $w = 1$, which gives $\kappa_n = 1$ and the effective-strength index $n\mu_n^2$.

For the oracle score-inversion result, [definition 14](#) is an Anderson–Rubin style inversion of the transported residual moment, in the tradition of [Anderson and Rubin \[1949\]](#) and [Fieller \[1954\]](#). At the true value θ_T , the centered score has variance controlled by the oracle weight second moment and the overlap constant. The radius $L_\alpha \sqrt{\widehat{\kappa}_n/n}$ combines that variance control with a uniform probability bound, yielding the honesty assertion in [theorem 3](#). The expected-length calculation uses the same score inequality together with the effective-strength lower restriction $t_n \geq t_0$, producing the order $\min\{1, t_0^{-1/2}\}$.

The finite-cell arguments show that sample-learned weights preserve the oracle rate. For [theorem 6](#), the target empirical frequencies estimate target cell probabilities, the source cellwise averages estimate reduced-form and first-stage contrasts, and the U-statistic $\hat{\kappa}_{U,n}$ estimates the collision scale associated with target-to-source weight dispersion. The sub-root cell-growth condition ensures that the additional sampling variation from cell learning remains on the same scale as the oracle score radius. The regular-cell extension in [theorem 7](#) repeats this control with known source-cell masses $q_{x,n}$ and known cell-varying propensity, using the regularity constants c_- and c_+ to keep the cellwise variance terms uniformly bounded.

Verification note. The top-level theorem statements and proof obligations corresponding to [theorems 1 to 7](#) are checked in Lean 4. The auxiliary notation objects [definitions 1, 4](#) and [11 to 13](#) organize the manuscript notation used by those checked declarations, and [definition 23](#) records the checked fixed-geometry separation calculation used in the lower-bound proofs.

Proof of [theorem 1](#). Write

$$H(O) = \begin{cases} 1/e(X), & Z = 1, \\ -1/\{1 - e(X)\}, & Z = 0. \end{cases}$$

1. Model membership in $\mathcal{P}_n$, as specified in [definition 3](#), supplies the full-data support, population presence, two-sample laws, overlap, source observation, randomization, exclusion, monotonicity, transport, target-complier positivity, domination, and envelope facts used below. The source observation map $(S, X, D(0), D(1), Y(0), Y(1), Z) \mapsto (X, Z, D(Z), Y(D(Z)))$ is measurable, and its X -marginal agrees with the source full-data X -marginal. Therefore overlap transfers to the observed source law:

$$\varepsilon \leq e(X) \leq 1 - \varepsilon \quad P_S\text{-a.s.}$$

Consequently H is measurable and $|H| \leq 1/\varepsilon$ P_S -almost surely. Since the observed outcome lies in $[0, 1]$ and the receipt indicator is bounded by one, both $H(O)Y$ and $H(O)D$ are integrable under P_S .

2. The inverse-propensity calculation used for the full-data contrasts is as follows. Let F_0, F_1 be measurable and integrable under $P_n^F(\cdot \mid S = 1)$, and assume that the assigned-source inverse-propensity integrand displayed below is integrable. For each measurable A , conditional randomization and overlap identify the $Z = z$ slice of the assigned source law, mapped back to full data, as the source full-data law weighted by $e(X)$ when $z = 1$ and by $1 - e(X)$ when $z = 0$. Splitting the assigned law into the two slices gives

$$\int_{\{X \in A\}} \left[\frac{\mathbf{1}\{Z = 1\}}{e(X)} F_1 - \frac{\mathbf{1}\{Z = 0\}}{1 - e(X)} F_0 \right] dP_{\text{assign}, S} = \int_{\{X \in A\}} (F_1 - F_0) dP_n^F(\cdot \mid S = 1).$$

On the $Z = 1$ slice the density $e(X)$ cancels $1/e(X)$, and on the $Z = 0$ slice the density $1 - e(X)$ cancels $-1/\{1 - e(X)\}$; the two slice integrals then add to the displayed contrast.

3. The chosen version Δ_Y is measurable, and the source-observation facts give, for every measurable covariate set A ,

$$\int_{\{O: X \in A\}} H(O)Y dP_S(O) = \int_A \Delta_Y(x) dP_S^X(x).$$

For the source full-data comparison used in the transport step, apply the preceding identity with $F_z = Y(D(z))$. The required integrability of the assigned-source inverse-propensity integrand follows by pulling back the already established integrability of $H(O)Y$ through the source observation map. Pulling the observed integral back through that map and splitting over $Z, D(0), D(1)$ gives

$$\int_{\{O: X \in A\}} H(O)Y \, dP_S(O) = \int_{\{X \in A\}} \{Y(D(1)) - Y(D(0))\} \, dP_n^F(\cdot \mid S = 1).$$

Exclusion rewrites $Y(D(z))$ as $Y^Z(z)$, and the source outcome-transport identity rewrites the last display as the integral over A of the source conditional assignment-outcome contrast. Hence Δ_Y agrees P_S^X -almost surely with that source contrast, while the displayed source-observation identity is the required observed source representation.

4. The chosen version Δ_D is measurable, and the source-observation facts give, for every measurable covariate set A ,

$$\int_{\{O: X \in A\}} H(O)D \, dP_S(O) = \int_A \Delta_D(x) \, dP_S^X(x).$$

For the source full-data comparison, apply the inverse-propensity identity with $F_z = D(z)$, written as the real indicator of receipt. The required integrability of the assigned-source inverse-propensity integrand follows by pulling back the already established integrability of $H(O)D$ through the source observation map. Pulling the observed integral back and splitting over $Z, D(0), D(1)$ gives

$$\int_{\{O: X \in A\}} H(O)D \, dP_S(O) = \int_{\{X \in A\}} \{D(1) - D(0)\} \, dP_n^F(\cdot \mid S = 1).$$

The source receipt-transport identity rewrites the last display as the integral over A of the source conditional receipt contrast. Hence Δ_D agrees P_S^X -almost surely with that source contrast, while the displayed source-observation identity is the required observed source representation.

5. For the transported reduced-form outcome mean, the already proved source almost-sure equality between Δ_Y and the source conditional assignment-outcome contrast first rewrites the integrand in $\mu_{Y,n}$. The Radon–Nikodym transport identity and target domination then change measure from P_S^X to P_T . The target outcome-transport equality replaces the transported source contrast by the target contrast, and the target full-data integral identity gives

$$\mu_{Y,n} = \int \{Y^Z(1) - Y^Z(0)\} \, dP_n^F(\cdot \mid S = 0).$$

Exclusion changes $Y^Z(z)$ to $Y(D(z))$. Finally, monotonicity permits a pointwise case split over $D(0), D(1)$, yielding

$$\mu_{Y,n} = \int \{Y(1) - Y(0)\} \mathbf{1}\{D(1) = 1, D(0) = 0\} \, dP_n^F(\cdot \mid S = 0).$$

6. For the transported first-stage mean, the weighted source representation first unfolds μ_n as

$$\mu_n = \int w(x) \Delta_D(x) dP_S^X(x).$$

The envelope $0 \leq w \leq 2k_n$, the bound $|H| \leq 1/\varepsilon$, and the receipt-indicator bound give integrability of the weighted observed receipt-score integrand. Separately, the envelope and $0 \leq \Delta_D \leq 1$ give integrability of $w\Delta_D$. Using the source almost-sure equality between Δ_D and the source receipt contrast, Radon–Nikodym transport, the target receipt-transport equality, and then the target full-data receipt identity,

$$\mu_n = \int \{D(1) - D(0)\} dP_n^F(\cdot \mid S = 0).$$

Monotonicity turns $D(1) - D(0)$ into $\mathbf{1}\{D(1) = 1, D(0) = 0\}$, and integration of this indicator gives

$$\mu_n = P_n^F\{D(1) = 1, D(0) = 0 \mid S = 0\}.$$

7. By the definition of θ_T , the numerator is the target complier-weighted outcome contrast and the denominator is the target complier share. Substituting the two identities just proved gives

$$\frac{\mu_{Y,n}}{\mu_n} = \theta_T.$$

8. Let $\mu = P_n^F(\cdot \mid S = 0)$ and $C = \{D(1) = 1, D(0) = 0\}$. Full-data support makes P_n^F a probability law, and together with population presence for $S = 0$ it makes μ a probability law. Target-complier positivity gives $0 < \mu(C)$. The conditional law μ is absolutely continuous with respect to P_n^F , so bounded full-data support gives

$$-1 \leq Y(1) - Y(0) \leq 1 \quad \mu\text{-a.s.}$$

The set C is measurable, so integral monotonicity gives

$$-\mu(C) \leq \int (Y(1) - Y(0)) \mathbf{1}_C d\mu \leq \mu(C).$$

Dividing by the positive number $\mu(C)$ yields $\theta_T \in [-1, 1] = \Theta$.

□

Proof of [theorem 4](#). Fix $t_0 > 0$. The assumptions on N_n , k_n , ε , and the admissible geometry imply that, for all sufficiently large n , the fixed-geometry slice contains a law with effective strength at least t_0 . On those indices, the liminf coverage term is the ordinary slice infimum, and the strength-restricted risk supremum is over a nonempty set.

1. For the lower bound, set

$$\rho = \frac{1 - \alpha}{8}, \quad H = \min\{1/4, \rho t_0^{-1/2}\}.$$

Then $\rho > 0$ and $H > 0$. The fixed-geometry least-favorable construction supplies, for all sufficiently large n , a family $\{Q_{n,h}^g : |h| \leq H\}$ from [definitions 19](#) and [23](#). Each member lies

in the fixed-geometry slice of [definition 16](#); the centered member has effective strength t_0 ; the target contrast equals $1/2 + h$; and

$$\|Q_{n,h}^{\mathfrak{g}} - Q_{n,0}^{\mathfrak{g}}\|_{\text{TV}} \leq 2\rho.$$

2. Let $(C_n) \in \mathfrak{C}^{\text{or}}(\mathfrak{g})$ be any fixed-geometry honest oracle procedure. For each row define

$$a_n = \inf_{P \in \mathcal{P}_n(\mathfrak{g})} P\{\theta_T(P) \in C_n\}, \quad r_n = \sup_{\substack{P \in \mathcal{P}_n(\mathfrak{g}) \\ t_n(P) \geq t_0}} \mathbb{E}_P[\lambda(C_n)].$$

The coverage values lie in $[0, 1]$. Since every confidence set is contained in $\Theta = [-1, 1]$, every expected length is at most 2, and the strength-restricted risk terms are bounded between 0 and 2 for all sufficiently large n .

3. Fix a sufficiently large n and put $I = [1/2 - H, 1/2 + H]$. For $u \in I$, compare the law $Q_{n,u-1/2}^{\mathfrak{g}}$ to the centered law $Q_{n,0}^{\mathfrak{g}}$. Because the target contrast under $Q_{n,u-1/2}^{\mathfrak{g}}$ is u , fixed-geometry coverage gives

$$a_n \leq Q_{n,u-1/2}^{\mathfrak{g}}\{u \in C_n\}.$$

The total-variation bound gives the comparison with $Q_{n,0}^{\mathfrak{g}}$. Applying the coverage-to-expected-length reduction over the interval I , whose Lebesgue length is $2H$, yields

$$2H(a_n - 2\rho) \leq \mathbb{E}_{Q_{n,0}^{\mathfrak{g}}}[\lambda(C_n)] \leq r_n.$$

4. Pass the preceding inequality to limits. The honesty condition in [definition 17](#) gives

$$1 - \alpha \leq \liminf_{n \rightarrow \infty} a_n.$$

The affine map $x \mapsto 2H(x - 2\rho)$ is monotone, and the boundedness from Step 2 justifies the $\liminf$ – $\limsup$ comparison. Hence every honest (C_n) satisfies

$$2H(1 - \alpha - 2\rho) \leq \limsup_{n \rightarrow \infty} r_n = R_{\mathfrak{g}}((C_n), t_0).$$

5. The constants reduce by elementary arithmetic. If $\rho t_0^{-1/2} \leq 1/4$, then $H = \rho t_0^{-1/2}$ and

$$2H(1 - \alpha - 2\rho) = \frac{3(1 - \alpha)^2}{16} t_0^{-1/2}.$$

If $1/4 \leq \rho t_0^{-1/2}$, then $H = 1/4$, and the same expression dominates

$$\frac{3(1 - \alpha)^2}{16} \min\{1, t_0^{-1/2}\}.$$

Thus

$$c_{\alpha} \min\{1, t_0^{-1/2}\} \leq R_{\mathfrak{g}}((C_n), t_0)$$

for every $(C_n) \in \mathfrak{C}^{\text{or}}(\mathfrak{g})$. The procedure $C_n = \Theta$ has coverage one on every row, so the honest class is nonempty; taking the infimum in [definition 18](#) gives the lower bound for $V_{\mathfrak{g}}^*(t_0)$.

6. For the upper bound, put

$$L = L_{\alpha} = \left(\frac{8}{\alpha \varepsilon^2} \right)^{1/2}$$

and define an oracle score-inversion rule using the fixed geometry: for every sufficiently large index, take the set in [definition 14](#) with w equal to the Radon–Nikodym weight of the target covariate law relative to the source covariate law in $\mathfrak{g}$, and with propensity e equal to the fixed propensity in $\mathfrak{g}$. Choose any measurable value at the finitely many earlier indices; changing finitely many values leaves the limiting statements unchanged. Measurability and containment in Θ follow from the measurability of the fixed weight and propensity and from the construction of the inversion set.

7. Fix $n > 0$ and $P \in \mathcal{P}_n(\mathfrak{g})$. The slice condition identifies the oracle weight and propensity with those of P , up to the admissible almost-sure weight version. The compact range result in [theorem 1](#) gives $\theta_T \in \Theta$, identifies the transported first stage with the target complier share, and gives

$$\frac{\mu_{Y,n}}{\mu_n} = \theta_T, \quad \mu_n > 0.$$

The Radon–Nikodym change-of-measure identity gives $\mathbb{E}_S[w(X)] = 1$, so the Kish dispersion satisfies $\kappa_n \geq 1$. A conditional-expectation pullout identity converts weighted source-score integrals into the corresponding covariate integrals. Applying it to the receipt score gives mean μ_n , while instrument overlap bounds its variance by κ_n/ε^2 . Chebyshev’s inequality therefore yields

$$P\left(\mu_n/2 < \left|\widehat{B}_n - \mu_n\right|\right) \leq \frac{4}{\varepsilon^2 t_n}.$$

The generic affine-inversion length bound then gives the pointwise inequality

$$\mathbb{E}_P[\lambda(C_n)] \leq \max\left\{2, 4L + \frac{8}{\varepsilon^2}\right\} \min\{1, t_n^{-1/2}\}.$$

Changing finitely many initial indices leaves the limiting expected-length bound unchanged.

8. Define

$$\delta_n = \begin{cases} 1, & n = 0, \\ 16k_n^2/n, & n > 0. \end{cases}$$

Since $k_n/\sqrt{n} \rightarrow 0$, we have $\delta_n \rightarrow 0$.

9. The same fixed row $n > 0$ also gives the coverage bound. For the true θ_T , the transported residual score has mean zero and variance at most $4\kappa_n/\varepsilon^2$. The empirical Kish statistic has mean κ_n , and the envelope $0 \leq w \leq 2k_n$ gives

$$P(\widehat{\kappa}_n < \kappa_n/2) \leq \frac{16k_n^2}{n\kappa_n} \leq \delta_n.$$

With

$$a = L\sqrt{\frac{\kappa_n}{2n}},$$

the identity $L^2 = 8/(\alpha\varepsilon^2)$ and Chebyshev’s inequality give

$$P\left(\left|\widehat{A}_n - \theta_T \widehat{B}_n\right| > a\right) \leq \alpha.$$

On the complement of these two bad events,

$$\left|\widehat{A}_n - \theta_T \widehat{B}_n\right| \leq L\sqrt{\widehat{\kappa}_n/n},$$

so $\theta_T \in C_n$. Hence, for every $P \in \mathcal{P}_n(\mathfrak{g})$,

$$1 - \alpha - \delta_n \leq P\{\theta_T(P) \in C_n\}.$$

Changing the value assigned at $n = 0$ leaves the limiting coverage statement unchanged.

10. The coverage values lie in $[0, 1]$, the slice is nonempty for all sufficiently large n , and $\delta_n \rightarrow 0$. The preceding coverage inequality therefore implies

$$(C_n) \in \mathfrak{C}^{\text{or}}(\mathfrak{g})$$

in the sense of [definition 17](#).

11. Let

$$C_0 = \max \left\{ 2, 4L + \frac{8}{\varepsilon^2} \right\}.$$

The pointwise length inequality in Step 7 and the strength restriction $t_n \geq t_0$ imply, row by row and then after taking the limsup frontier risk in [definition 7](#),

$$R_{\mathfrak{g}}((C_n), t_0) \leq C_0 \min\{1, t_0^{-1/2}\}.$$

Since (C_n) is admissible for the infimum in [definition 18](#), and the risk functional is nonnegative by compactness of Θ , this gives

$$V_{\mathfrak{g}}^*(t_0) \leq C_0 \min\{1, t_0^{-1/2}\}.$$

Substituting $L = L_{\alpha}$ makes $C_0 = C_{\alpha}$.

Combining Steps 5 and 11 gives the asserted two-sided bound. The constants used in both arguments depend only on α and ε , so they are uniform over all admissible deterministic geometries $\mathfrak{g}$. $\square$

Proof of [theorem 5](#). Fix the admissible geometry array $\mathfrak{g}$. We prove the three asserted conclusions in the order in which they appear.

1. For each n , the unit-weight hypothesis gives

$$w_n(x)^2 = 1 \quad P_{S,n}^X\text{-a.s.}$$

Hence, by the definition of the geometry second moment and because $P_{S,n}^X$ is a probability law under [definition 15](#),

$$\kappa_n = \int w_n(x)^2 dP_{S,n}^X(x) = \int 1 dP_{S,n}^X = 1.$$

For the equality of covariate laws, let $A \subseteq \mathcal{X}$ be measurable. The change-of-measure clause in [definition 15](#) gives

$$P_{T,n}(A) = \int_A w_n(x) dP_{S,n}^X(x).$$

The same almost-sure equality $w_n = 1$ remains valid after restricting $P_{S,n}^X$ to A , so

$$P_{T,n}(A) = \int_A 1 dP_{S,n}^X = P_{S,n}^X(A).$$

Since this holds for every measurable A , the measures agree:

$$P_{T,n} = P_{S,n}^X.$$

2. Let $P \in \mathcal{P}_n(\mathfrak{g})$. By [definition 16](#), its source covariate marginal is $P_{S,n}^X$ from $\mathfrak{g}$, and its Radon–Nikodym transport weight agrees $P_{S,n}^X$ -almost surely with the geometry weight w_n . Therefore the law-induced Kish dispersion of P equals the geometry second moment:

$$\int \{w_n^P(x)\}^2 dP_{S,n}^X(x) = \int w_n(x)^2 dP_{S,n}^X(x) = \kappa_n.$$

By the first step, this common value is 1. Substituting this into the definition of effective strength in [definition 7](#) yields

$$t_n = \frac{n\mu_n^2}{\kappa_n} = \frac{n\mu_n^2}{1} = n\mu_n^2.$$

3. Fix $t_0 > 0$. The hypotheses on c , N_n/n , ε , α , k_n , and the admissibility of $\mathfrak{g}$ are exactly the structural hypotheses required by [theorem 4](#). For every $P \in \mathcal{P}_n(\mathfrak{g})$, the defining membership condition in [definition 16](#) contains the transported-IV model assumptions needed there: the two-sample array condition, instrument overlap, the weight envelope, the weight second-moment condition, and the degrading-array condition. Applying [theorem 4](#) therefore gives, with

$$L_\alpha = \sqrt{\frac{8}{\alpha\varepsilon^2}}, \quad c_\alpha = \frac{3(1-\alpha)^2}{16}, \quad C_\alpha = \max \left\{ 2, 4L_\alpha + \frac{8}{\varepsilon^2} \right\},$$

the bounds

$$c_\alpha \min\{1, t_0^{-1/2}\} \leq V_{\mathfrak{g}}^*(t_0) \leq C_\alpha \min\{1, t_0^{-1/2}\}.$$

□

Proof of [theorem 2](#). 1. Set

$$c_0 = \frac{3(1-\alpha)^2}{16}, \quad t_c = 1.$$

Since $0 < \alpha < 1$, $1 - \alpha > 0$, and hence $c_0 > 0$; also $t_c > 0$. Fix an arbitrary measurable covariate space, arbitrary arrays N_n, k_n satisfying the displayed size and cell-scale assumptions, and a threshold $t_0 > 0$.

2. Choose the measurable singleton carriers $x_{n,i}$, $i \in \{1, \dots, k_n\}$, supplied by the carrier assumption. Let

$$\mu_n = \sum_{i=1}^{k_n} k_n^{-1} \delta_{x_{n,i}}.$$

The positivity of k_n gives $k_n k_n^{-1} = 1$, so each μ_n is a probability measure. Define the deterministic geometry

$$P_{S,n}^X = \mu_n, \quad P_{T,n} = \mu_n, \quad w_n \equiv 1, \quad e_n \equiv \frac{1}{2}.$$

The probability, overlap, envelope, second-moment, and domination requirements in [definition 15](#) follow directly from these definitions and from $0 < \varepsilon < 1/2$, with the change-of-measure identity reducing to integration of the constant weight 1.

3. The fixed-geometry slice associated with this $\mathbf{g}$ is contained in the finite-cell class $\mathcal{N}_n$ of [definition 22](#) at each n . Indeed, the source covariate law is the uniform measure on the displayed k_n carriers, the carrier cells are measurable singletons, each cell has source mass $1/k_n$, the target law agrees with the source law, and the transport weight is 1 almost surely.
4. Consider the oracle procedure D^{full} that always returns the whole compact parameter space $\Theta = [-1, 1]$. Its graph is measurable, its values are subsets of Θ , and it is invariant to almost-sure changes of the supplied weight. For every transported-IV row P , [theorem 1](#) places $\theta_T(P, n)$ in Θ , so the coverage of D^{full} is exactly one. Taking the infimum over the nonempty transported class preserves the value one, and therefore D^{full} is oracle-honest in the sense of [definition 6](#). Thus the class over which $V^*(t_0)$ and $V_{\mathcal{N}}^*(t_0)$ take their infima is nonempty.
5. On the unit-weight geometry just constructed, the weight is 1 almost surely for every n . Applying [theorem 5](#) yields, for every $t_0 > 0$,

$$c_0 \min\{1, t_0^{-1/2}\} \leq V_{\mathbf{g}}^*(t_0),$$

where $V_{\mathbf{g}}^*(t_0)$ is the fixed-geometry value from [definition 18](#).

6. Let C be any oracle-honest procedure. Restricting C to the fixed geometry gives a fixed-geometry honest procedure, and the global frontier risk from [definition 7](#) dominates the corresponding fixed-geometry risk. Hence

$$V_{\mathbf{g}}^*(t_0) \leq R((C_n), t_0).$$

Taking the infimum over all oracle-honest C in [definition 8](#) gives

$$c_0 \min\{1, t_0^{-1/2}\} \leq V^*(t_0).$$

7. For the finite-cell value, first note the two boundedness facts used in the limsup comparison. The length criterion is always between 0 and 2, since it is Lebesgue length after intersecting with $\Theta = [-1, 1]$. Therefore every fixed-geometry risk term is nonnegative: if the strength-indexed slice is empty the supremum is zero, and otherwise the expected length of any selected law is nonnegative. The same length bound gives an eventual upper bound by 2 for the relevant fixed and finite-cell risk terms.
8. Fix an arbitrary oracle-honest procedure C . Build a fixed-geometry oracle procedure D by feeding C the observed sample together with the canonical Radon–Nikodym weight $dP_{T,n}/dP_{S,n}^X$ of $\mathbf{g}$ and the fixed propensity e_n . On every row $P \in \mathcal{P}_n(\mathbf{g})$, the source law, target law, weight, and propensity agree with those inputs, so D 's confidence set, coverage probability, and expected length agree with those of C on that row. For all sufficiently large n , the fixed-geometry strength slice is nonempty, which supplies the required coverage comparison; the coverage probabilities lie between 0 and 1, so the liminf comparison transfers oracle honesty of C to fixed-geometry honesty of D .
9. For all sufficiently large n , the fixed-geometry strength slice has admissible laws. On such rows,

$$\sup_{P \in \mathcal{P}_n(\mathbf{g}): t_n \geq t_0} \mathbb{E}_P \lambda(D_n) \leq \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P \lambda(C_n),$$

because $\mathcal{P}_n(\mathbf{g}) \subseteq \mathcal{N}_n$ and the expected lengths of D and C agree on the slice. The nonnegativity and boundedness from the previous step allow the limsup comparison, hence the fixed-geometry risk of D is bounded by the finite-cell oracle risk of C . Taking the infimum over oracle-honest C , and using the fixed-geometry lower bound, gives

$$c_0 \min\{1, t_0^{-1/2}\} \leq V_{\mathcal{N}}^*(t_0).$$

10. Combining the preceding inequalities proves the two displayed lower bounds for every $t_0 > 0$. If $0 < t_0 \leq t_c = 1$, then $t_0^{-1/2} \geq 1$, so

$$\min\{1, t_0^{-1/2}\} = 1,$$

and the oracle lower bound yields $c_0 \leq V^*(t_0)$. The constants c_0 and t_c were chosen before the covariate space, arrays, transported laws, and effective-strength decomposition; the argument uses a row only through the threshold condition $t_n = n\mu_n^2/\kappa_n \geq t_0$. Thus the same constants apply uniformly over all decompositions of t_0 into μ_n and κ_n . $\square$

Proof of [theorem 3](#). 1. **Construction.** Choose an admissible geometry $\mathbf{g}$ supplied by the hypothesis. For every $t_0 > 0$, the fixed-geometry construction supplies admissible comparison laws for all sufficiently large n , and hence the full class $\mathcal{P}_n$ is nonempty at those indices. Set

$$L_\alpha = \left(\frac{8}{\alpha\varepsilon^2}\right)^{1/2}, \quad C_0 = \max\left\{2, 4L_\alpha + \frac{8}{\varepsilon^2}\right\}.$$

Since $C_0 \geq 2$, $C_0 > 0$. Define C_n by the score-inversion construction in [definition 14](#), using the oracle weight and propensity as inputs. Thus, for every source sample,

$$C_n = \left\{\theta \in \Theta : \left|\hat{A}_n - \theta\hat{B}_n\right| \leq L_\alpha\sqrt{\hat{\kappa}_n/n}\right\}.$$

The range condition follows from the defining intersection with Θ . The graph is measurable because $\hat{A}_n$, $\hat{B}_n$, and $\hat{\kappa}_n$ are measurable functions of the source sample whenever the supplied weight and propensity are measurable. If two versions of the oracle weight agree almost surely under P_S^X , then the corresponding inversion sets agree almost surely under the two-sample law.

2. **Weighted conditional-expectation reduction.** The proof uses the following pull-out identity. Suppose F is a source-observation function, d is a covariate function, and w is a measurable transport weight. If, for every measurable A ,

$$\int_{\{o: X(o) \in A\}} F(o) dP_S(o) = \int_A d(x) dP_S^X(x),$$

then

$$\int w(X(o))F(o) dP_S(o) = \int w(x)d(x) dP_S^X(x).$$

Indeed, the set-integral identity identifies $d(X)$ with the conditional expectation of F given X ; multiplying by the X -measurable function $w(X)$ and integrating gives the displayed equality. The measurability and integrability requirements are supplied by overlap, the weight envelope, and the bounded source-observation variables.

3. **Pointwise expected length.** Fix $n > 0$ and $P \in \mathcal{P}_n$. The compact causal-range conclusion in [theorem 1](#) gives $\theta_n = \theta_T(P, n) \in \Theta$, identifies the transported first stage with the target complier share, and therefore gives $\mu_n > 0$. The Radon–Nikodym weight has mean one under P_S^X , so its second moment κ_n satisfies $\kappa_n \geq 1$.

Let

$$B_n = \frac{1}{n} \sum_{i=1}^n w(X_i) H_i D_i.$$

The score identities and the pull-out identity give $\mathbb{E}_P B_n = \mu_n$. The overlap bound gives

$$\text{Var}_P(B_n) \leq \frac{\kappa_n}{\varepsilon^2 n},$$

and hence

$$\Pr_P \left\{ \frac{\mu_n}{2} < |B_n - \mu_n| \right\} \leq \frac{4}{\varepsilon^2 t_n}, \quad t_n = \frac{n \mu_n^2}{\kappa_n}.$$

The empirical Kish statistic $\hat{\kappa}_n = n^{-1} \sum_i w(X_i)^2$ is nonnegative and has mean κ_n . Applying the affine score-inversion length bound to $\hat{A}_n$, $\hat{B}_n$, and $\hat{\kappa}_n$ gives

$$\mathbb{E}_P \lambda(C_n) \leq C_0 \min\{1, t_n^{-1/2}\}.$$

4. **Pointwise coverage.** Define

$$\delta_n = \begin{cases} 1, & n = 0, \\ 16k_n^2/n, & n > 0. \end{cases}$$

The scale condition $k_n/\sqrt{n} \rightarrow 0$ gives $\delta_n \rightarrow 0$. Changing finitely many initial indices leaves the limiting coverage requirement unchanged. For every sufficiently large n , set

$$S_n = \hat{A}_n - \theta_n \hat{B}_n, \quad K_n = \hat{\kappa}_n,$$

and use the ratio identity from [theorem 1](#). The pull-out identity gives mean zero for the affine score $w(X_i) H_i (Y_i - \theta_n D_i)$, and overlap and boundedness give

$$\text{Var}_P(S_n) \leq \frac{4\kappa_n}{\varepsilon^2 n}.$$

The envelope $0 \leq w \leq 2k_n$ gives the empirical-Kish lower-tail bound

$$\Pr_P \{K_n < \kappa_n/2\} \leq \frac{16k_n^2}{n\kappa_n} \leq \frac{16k_n^2}{n}.$$

With

$$a_n = L_\alpha \sqrt{\frac{\kappa_n}{2n}},$$

the identity $L_\alpha^2 = 8/(\alpha\varepsilon^2)$ and Chebyshev's inequality yield

$$\Pr_P \{|S_n| > a_n\} \leq \alpha.$$

On the complement of these two bad events,

$$|S_n| \leq a_n \leq L_\alpha \sqrt{K_n/n},$$

and $\theta_n \in \Theta$, so $\theta_n \in C_n$. Therefore, uniformly over $P \in \mathcal{P}_n$,

$$1 - \alpha - \delta_n \leq \Pr_P \{\theta_n \in C_n\}.$$

5. **Oracle honesty.** Coverage probabilities lie in $[0, 1]$, the class is nonempty for all sufficiently large n , and $\delta_n \rightarrow 0$. On those indices, the infimum over $P \in \mathcal{P}_n$ agrees with the oracle coverage infimum for the constructed score-inversion procedure. Taking the liminf of these infima in [definition 6](#) gives

$$1 - \alpha \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}_n} \Pr\{\theta_T(P, n) \in C_n\}.$$

Thus $(C_n) \in \mathfrak{C}^{\text{or}}$.

6. **Frontier risk of the constructed sequence.** Fix $t_0 > 0$. Changing finitely many initial indices leaves the limiting risk unchanged, and every sufficiently large n uses the pointwise expected-length bound above. Since $t \mapsto \min\{1, t^{-1/2}\}$ is decreasing on $(0, \infty)$, every row with $t_n \geq t_0$ satisfies

$$\mathbb{E}_P \lambda(C_n) \leq C_0 \min\{1, t_0^{-1/2}\}.$$

Taking the supremum over such P and then the limsup in [definition 7](#) gives

$$R((C_n), t_0) \leq C_0 \min\{1, t_0^{-1/2}\}.$$

7. **Oracle value.** For every admissible geometry $\mathfrak{g}' \in \mathcal{G}$, [theorem 4](#) gives

$$\frac{3(1 - \alpha)^2}{16} \min\{1, t_0^{-1/2}\} \leq V_{\mathfrak{g}'}^*(t_0) \leq C_0 \min\{1, t_0^{-1/2}\}.$$

Using the selected geometry $\mathfrak{g}$, every globally oracle-honest procedure induces a fixed-geometry honest procedure, so the same lower bound applies to the global oracle risk of any $(E_n) \in \mathfrak{C}^{\text{or}}$. Since the constructed (C_n) is oracle honest, the infimum defining $V^*(t_0)$ is over a nonempty class. Also, $\lambda(A) \leq \lambda(\Theta) = 2$ for every reported set A , so the relevant suprema are uniformly bounded for the limsup comparison. Hence the risks over oracle-honest procedures are bounded below, and evaluating the infimum at the constructed sequence gives

$$V^*(t_0) \leq C_0 \min\{1, t_0^{-1/2}\}.$$

Combining the displayed bounds proves the asserted frontier-risk bound and the corresponding attainable oracle-value order for every fixed $t_0 > 0$. □

Proof of [theorem 7](#). 1. The measurable-carrier hypothesis and the regular-cell mass bounds give a member of $\mathcal{N}_n^{\text{reg}}$ in every row n , so the coverage term in $\mathfrak{C}^{\text{reg}}$ is evaluated by an ordinary infimum. Put

$$B_{\varepsilon, c} = 8(\varepsilon^{-2} + c^{-1}), \quad C_0 = \max\{2, 4\sqrt{2}L + 8B_{\varepsilon, c}\}.$$

Then $B_{\varepsilon, c} \geq 0$, $L \geq 0$, and $C_0 > 0$. Define C_n^{reg} by the regular-cell inversion rule, with the stated Θ -valued convention when $N_n < 2$. The rule is the measurable regular-cell procedure obtained by applying the known source masses $q_{x, n}$, the known propensity values, and the source and target samples to the score inversion map.

2. For every $P \in \mathcal{N}_n^{\text{reg}}$, the constructed random set agrees almost surely with the regular-cell inversion rule using the ambient source-cell mass function and propensity score. This is the representation used in the honesty argument. On rows with $2 \leq N_n$, that same inversion rule is the affine set

$$\left\{ \vartheta \in \Theta : |\widehat{M}_{Y,n} - \vartheta \widehat{M}_{D,n}| \leq L \sqrt{\widehat{K}_n/n} \right\},$$

whereas on rows with $N_n < 2$ it is Θ by definition. Since the class is nonempty in every row, the coverage term in [definition 21](#) is

$$\inf_{P \in \mathcal{N}_n^{\text{reg}}} P(\theta_T \in C_n^{\text{reg}}).$$

The uniform coverage bound for the regular-cell inversion rule applies at the critical inequality $L \geq \{2B_{\varepsilon,c}/\alpha\}^{1/2}$, and yields

$$1 - \alpha \leq \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{N}_n^{\text{reg}}} P(\theta_T \in C_n^{\text{reg}}).$$

Thus $(C_n^{\text{reg}}) \in \mathfrak{C}^{\text{reg}}$.

3. Fix $t_0 > 0$. From $N_n/n \rightarrow c > 0$, eventually $n > 0$ and $2 \leq N_n$. On this tail, for every $P \in \mathcal{N}_n^{\text{reg}}$, the compact causal-range identities give a positive transported first stage μ_n . The regular-cell moment calculations give integrability of the outcome and receipt score moments, nonnegativity of $\widehat{K}_n$, and

$$\mathbb{E}_P[\widehat{K}_n] = 1 + \kappa_n \leq 2\kappa_n.$$

The first-stage deviation event satisfies

$$P\{|\widehat{M}_{D,n} - \mu_n| > \mu_n/2\} \leq \frac{4B_{\varepsilon,c}}{t_n}, \quad t_n = \frac{n\mu_n^2}{\kappa_n}.$$

Using the almost-sure representation from the preceding step together with the nondegenerate identity for the regular-cell inversion rule, the expected length of C_n^{reg} is the expected length of this affine inversion set for all sufficiently large n . Applying the inflated affine-inversion length bound with $Y = 8B_{\varepsilon,c}$ gives, uniformly over $P \in \mathcal{N}_n^{\text{reg}}$,

$$\mathbb{E}_P[\lambda(C_n^{\text{reg}})] \leq C_0 \min\{1, t_n^{-1/2}\}$$

on that tail. The abstract frontier-risk monotonicity then converts the pointwise bound into

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n^{\text{reg}}: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n^{\text{reg}})] \leq C_0 \min\{1, t_0^{-1/2}\}.$$

4. For the lower bound, choose the injected singleton carrier in each row and put the same uniform law on the source and target covariates:

$$\nu_n = \sum_{i=1}^{k_n} k_n^{-1} \delta_{x_{i,n}}, \quad w_n \equiv 1, \quad e_n \equiv \frac{1}{2}.$$

This deterministic geometry is admissible: ν_n is a probability measure supported on the chosen cells, the propensity obeys the overlap interval, the weight envelope and second-moment bounds hold, and the equal cell masses satisfy

$$\frac{c_-}{k_n} \leq \frac{1}{k_n} \leq \frac{c_+}{k_n}.$$

Consequently, every fixed-geometry slice for this geometry is contained in $\mathcal{N}_n^{\text{reg}}$. By [theorem 4](#), its fixed-geometry honest minimax risk is bounded below by

$$\frac{3(1-\alpha)^2}{16} \min\{1, t_0^{-1/2}\}.$$

5. Let $C \in \mathfrak{C}^{\text{reg}}$ be arbitrary. Construct a fixed-geometry oracle procedure D by feeding C the observed two-sample draw, the selected regular-cell design, the mass vector $q_{x,n} = 1/k_n$, and the propensity vector $e_{x,n} = 1/2$ whenever the sample lies on the selected support; on the complementary event set $D_n = \Theta$. Under every law in the fixed-geometry slice, the support event has probability one, so D_n and C_n agree almost surely after the slice is viewed as a regular-cell law. Hence their coverages agree on the slice, and on the tail where the fixed-geometry slice has admissible laws, the fixed-geometry coverage infimum is at least the corresponding regular-cell infimum. Thus D is fixed-geometry honest. The same almost-sure equality gives equality of expected lengths on the slice, while every set length is bounded by $\lambda(\Theta) = 2$; therefore the fixed-geometry risk of D is at most the regular-cell risk of C . Combining this domination with the previous fixed-geometry lower bound and then taking the infimum over all $C \in \mathfrak{C}^{\text{reg}}$ proves

$$\frac{3(1-\alpha)^2}{16} \min\{1, t_0^{-1/2}\} \leq \inf_{C \in \mathfrak{C}^{\text{reg}}} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n^{\text{reg}}: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)].$$

Together with the constructed upper bound, this gives the stated regular-cell attainment conclusion. □

Proof of [theorem 6](#). 1. First record that every finite-cell row class contains an admissible law. The carrier condition, the sampling-ratio limit, the positivity and growth conditions on k_n , and the overlap bounds give an element of $\mathcal{N}_n$ for each n . Define

$$B_c = 32(1 + c^{-1}), \quad L_{\alpha,c} = \left(\frac{2B_c}{\alpha} \right)^{1/2}, \quad C_0 = \max\{2, 4\sqrt{2}L_{\alpha,c} + 8B_c\}.$$

To compare with the regular-cell construction, set

$$x = 4 + \frac{3}{c}, \quad a = x^{-1/2}, \quad \bar{\varepsilon} = \frac{1}{4} + \frac{1}{2}a.$$

Since $c > 0$, $x > 4$, hence $\sqrt{x} > 2$, $0 < a < 1/2$, and $0 < \bar{\varepsilon} < 1/2$. Moreover

$$\bar{\varepsilon}\sqrt{x} \geq 1, \quad \bar{\varepsilon}^{-2} \leq x.$$

With

$$B = 8(\bar{\varepsilon}^{-2} + c^{-1}),$$

this gives $0 \leq B \leq B_c$. Consequently

$$\left(\frac{2B}{\alpha}\right)^{1/2} \leq L_{\alpha,c},$$

and $C_0 > 0$ because $C_0 \geq 2$.

2. Apply [theorem 7](#) with overlap level $\bar{\varepsilon}$, regular-cell constants $c_- = c_+ = 1$, and critical constant $L_{\alpha,c}$. The hypotheses required there are supplied by the corresponding regular-cell row assumptions: the row supplies the two-sample condition, bounded overlap for its own propensity, and the degrading-array condition. The common propensity $1/2$ enters the uniform finite-cell argument through [assumption 16](#) and the reduction to the regular-cell design with $c_- = c_+ = 1$. The preceding bound on $L_{\alpha,c}$ supplies the required critical-value inequality. Thus there is a regular-cell procedure C^{reg} whose set agrees almost surely with the regular-cell score-inversion rule, which is honest, and whose risk is bounded by

$$\max\{2, 4\sqrt{2} L_{\alpha,c} + 8B\} \min\{1, t_0^{-1/2}\}$$

for every $t_0 > 0$.

3. Transport this regular-cell rule to the finite-cell experiment. On each finite-cell row, the finite-cell inversion rule, the ambient regular-cell rule, and the regular-cell procedure just obtained agree almost surely after identifying the uniform source-cell masses and the common propensity $1/2$. This reduction gives finite-cell honesty for the sample-only procedure

$$C_n(s) = \text{the finite-cell score-inversion set with critical constant } L_{\alpha,c}.$$

Equivalently, this is exactly the rule displayed in the theorem statement, with $C_n = \Theta$ when $N_n < 2$.

4. For the upper expected-length bound, use the regular-cell risk bound from Step 2, the risk comparison induced by the reduction to the regular-cell design, and the constant comparison $B \leq B_c$. Since

$$\min\{1, t_0^{-1/2}\} \geq 0,$$

$$\max\{2, 4\sqrt{2} L_{\alpha,c} + 8B\} \leq \max\{2, 4\sqrt{2} L_{\alpha,c} + 8B_c\} = C_0$$

implies

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)] \leq C_0 \min\{1, t_0^{-1/2}\}.$$

This proves the asserted attainment bound for every $t_0 > 0$.

5. For the converse bound, fix $t_0 > 0$. Choose, for every n , an injection from $\{1, \dots, k_n\}$ into the measurable carrier with measurable singleton image. In the deterministic geometry $\mathfrak{g}$, take both covariate laws to be

$$P_{S,n}^X = P_{T,n} = \sum_{i=1}^{k_n} k_n^{-1} \delta_{x_{i,n}},$$

take $w = 1$, and take $e = 1/2$. The cancellation $k_n k_n^{-1} = 1$ shows that these covariate laws are probability measures. The bounds $0 < \varepsilon < 1/2$ put the propensity inside the overlap interval, while $k_n \geq 1$ gives the weight envelope and second-moment requirements. Hence $\mathfrak{g} \in \mathcal{G}$.

6. For this geometry, the selected cell range is measurable, has both $P_{S,n}^X$ - and $P_{T,n}$ -probability one, and each chosen singleton has mass k_n^{-1} . Therefore any row in the fixed-geometry slice $\mathcal{P}_n(\mathbf{g})$ satisfies the finite-cell source condition with this cell injection, and the remaining model-class conditions are inherited from the fixed-geometry slice. Thus every such row lies in $\mathcal{N}_n$. The same measurable range also makes the event that all source and target sample covariates lie in the selected range a measurable sample event.
7. Invoke the lower-bound half of [theorem 4](#) for the geometry $\mathbf{g}$:

$$\frac{3(1-\alpha)^2}{16} \min\{1, t_0^{-1/2}\} \leq V_{\mathbf{g}}^*(t_0).$$

It is enough to show that $V_{\mathbf{g}}^*(t_0)$ is bounded above by the feasible finite-cell minimax value.

8. Take an arbitrary feasible finite-cell honest procedure $\tilde{C}$. Construct a fixed-geometry oracle procedure D as follows: on samples whose source and target covariates all lie in the selected cell range, set

$$D_n = \tilde{C}_n;$$

off that support, set $D_n = \Theta$. The graph measurability follows by splitting the sample space into the measurable support event and its complement; weight-invariance is immediate because this sample-and-support rule is fixed on each such event.

9. On every fixed-geometry row, the source and target covariate laws assign probability one to the selected range, so the product sampling law assigns probability one to the support event from Step 8. Hence D_n and $\tilde{C}_n$ agree almost surely on that row. Their coverage probabilities and expected lengths are therefore equal on fixed-geometry rows. For all sufficiently large n , the fixed-geometry threshold slice has admissible laws, and the finite-cell row class has admissible laws for every n . On those rows, the inclusion from Step 6 and the equality of coverage probabilities give

$$\inf_{P \in \mathcal{N}_n} \inf_P \Pr\{\theta_T \in \tilde{C}_n\} \leq \inf_{P \in \mathcal{P}_n(\mathbf{g})} \inf_P \Pr\{\theta_T \in D_n\}.$$

The coverage quantities are bounded below by 0, and the fixed-geometry coverage infimum is eventually bounded above by 1. Passing to $\liminf$ therefore preserves the displayed inequality, so finite-cell honesty of $\tilde{C}$ gives fixed-geometry oracle honesty of D .

10. For the risk comparison, first note that every set length is computed inside $\Theta = [-1, 1]$, so every realized length is at most 2. Hence, for $\tilde{C}$, each finite-cell expected length is at most 2. For all sufficiently large n , the fixed-geometry threshold slice has admissible laws. On those n , the equality of expected lengths on fixed-geometry rows and the inclusion from Step 6 give

$$\sup_{P \in \mathcal{P}_n(\mathbf{g}): t_n \geq t_0} \mathbb{E}_P[\lambda(D_n)] \leq \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(\tilde{C}_n)],$$

where the finite-cell supremum is controlled by the preceding length bound. The fixed-geometry supremum is eventually nonnegative because the threshold slice is nonempty and expected length is nonnegative. The compact parameter space bounds every finite-cell supremum by 2 under the stated supremum convention. These lower and upper bounds

supply the boundedness hypotheses for passing the eventual inequality through limsup, yielding

$$\text{fixed-geometry risk of } D \leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(\tilde{C}_n)].$$

11. It remains to place the fixed-geometry value below the risk of this particular D . The same length bound applies to every fixed-geometry oracle procedure: for any such procedure, every fixed-geometry expected length is at most 2. The supremum convention and nonnegativity of expected length keep each fixed-geometry row supremum between 0 and 2 on the relevant tail. These facts imply that every fixed-geometry oracle risk is nonnegative, so the collection of fixed-geometry honest risks is bounded below by 0. Since D is fixed-geometry honest by Step 9, the defining infimum for $V_g^*(t_0)$ satisfies

$$V_g^*(t_0) \leq \text{fixed-geometry risk of } D.$$

Combining this with Step 10 gives

$$V_g^*(t_0) \leq \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(\tilde{C}_n)].$$

Taking the infimum over all feasible finite-cell honest $\tilde{C}$ and combining with Step 7 yields

$$\frac{3(1-\alpha)^2}{16} \min\{1, t_0^{-1/2}\} \leq \inf_{\tilde{C} \in \mathfrak{C}^{\text{cell}}} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(\tilde{C}_n)].$$

This completes the proof. □

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